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MIL-M-28787/360(SH)
29 June 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

CONTROLLER, DATA MODULE, KEY CODE DFH

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic data controller module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbook. The following specification, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 1.2442 failures per million operating hours when predicted in accordance with DOD-HDBK-217E.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be DFH.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 140 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/ -3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	S	1	INF	1	INF	1	INF	MOhm
B1-B8	Continuity	III	X	0 0	1 0	0 0	1.0	0 0	1.0	Ohm
B9-B16	Continuity	III	Q	0 0	0.1	0 0	0.1	0 0	0.1	Ohm
C	Vcc stress	IV	S	----------Module shall pass all remaining tests----------						
D1-D40	Ibd	V	S	0 0	1 0	0 0	1.0	0 0	1.0	mA
D41-D60	Ibd	V	S	0 0	0.1	0 0	0.1	0 0	0.1	mA
D61,D62	Ibd	V	S	0.0	0 2	0 0	0.2	0 0	0.2	mA
D63	Ibd	V	S	0 0	2.0	0 0	2 0	0 0	2.0	mA
E1-E37	Vic	VI	S	1 5	-0 2	-1.5	-0.2	-1 5	-0.2	V
E38-E61	Vic	VI	S	-1 5	-0.4	-1 5	-0.4	-1 5	-0.4	V
E62,E63	Vic	VI	S	-1.2	-0.4	-1 2	-0.4	-1 2	-0.4	V
F1-F25	Iac	VII	S	-85	-25	-85	-25	-85	-25	mA
F26-F41	Ias	VII	S	-200	-60	-200	-60	-200	-60	mA
F42,F43	Ias	VII	S	-100	-40	-100	-40	-100	-40	mA
G1-G18	Iih	VIII	X	-70	70	-70	70	-70	70	uA
G17-G19	Iih	VIII	X	-295	-217	295	-217	-295	-217	uA
G20-G40	Iih	VIII	X	0	50	0	50	0	50	uA
G41-G60	Iih	VIII	X	0	20	0	20	0	20	uA
G61,G62	Iih	VIII	X	0	40	0	40	0	40	uA
G63	Iih	VIII	X	0	100	0	100	0	100	uA
H1-H16	Iil	IX	X	-410.0	0.0	-410.0	0.0	-410.0	0.0	uA
H17-H34	Iil	IX	X	-360.0	0.0	-360.0	0.0	-360 0	0.0	uA
H35-H37	Iil	IX	X	-386.0	-476.0	-386.0	-476.0	-386.0	-476.0	uA

- See footnotes at end of table

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TABLE I Module electrical requirements - Continued

Test conditions	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
H33-H59	Iil	IX	X	-400.0	0.0	-400.0	0.0	-400.0	0.0	uA
H60	Iil	IX	X	-3.2	0.0	-3.2	0.0	-3.2	0.0	mA
H61-H63	Iil	IX	X	-1.6	0.0	-1.6	0.0	-1.6	0.0	mA
J1-J25	Voh	X	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J26	Voh	X	X	3.0	4.5	3.0	4.5	3.0	4.5	V
J27	Voh	X	X	2.5	4.5	2.5	4.5	2.5	4.5	V
J28-J43	Voh	X	X	3.34	4.5	3.34	4.5	3.34	4.5	V
K1-K27	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K28-K43	Vol	XI	X	0.0	0.507	0.0	0.507	0.0	0.507	V
L1-L16	Iozh	XII	X	-70	70	-70	70	-70	70	uA
L17-L33	Iozh	XII	X	-50	50	-50	50	-50	50	uA
M1-M16	Iozl	XIII	X	-410	0	-410	0	-410	0	uA
M17-M33	Iozl	XIII	X	-50	50	-50	50	-50	50	uA
N	Icc	XIV	X	41	827	41	827	41	827	mA
P1,P2	Functional	XV	X	----------		See table XV		----------		
Q1,Q2	Dynamic	XVI	S	----------		See table XVI		----------		
Q3,Q4	Dynamic	XVI	Q	----------		See table XVI		----------		
R	PSFC	XVIII	S	3.6	6.6	3.6	6.6	3.6	6.6	uF
S	PSTA	XIX	Q	0	400	0	400	0	400	mV
T1-T43	Output crosstalk high level	XX	Q	0	500	0	500	0	500	mV

1/ See footnotes at end of table

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TABLE I. Module electrical requ rements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		0 + 5°C 4/		60 + 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
U1-U7	Output crosstalk, low level	XXI	Q	0	650	0	650	0	650	mV
U8-U41	Output crosstalk, low level	XXI	Q	0	500	0	500	0	500	mV
U42,U43	Output crosstalk, low level	XXI	Q	0	650	0	650	0	650	mV
V1-V21	Cin	XXII	Q	2	35	2	35	2	35	pF
V22-V27	Cin	XXII	Q	2	40	2	40	2	40	pF
V28	Cin	XXII	Q	4	40	4	40	4	40	pF
V29	Cin	XXII	Q	2	40	2	40	2	40	pF
V30	Cin	XXII	Q	4	40	4	40	4	40	pF
V31-V47	Cin	XXII	Q	2	35	2	35	2	35	pF
V48-V63	Cin	XXII	Q	2	40	2	40	2	40	pF
W1-W7	Cout	XXIII	Q	2	40	2	40	2	40	pF
X1-X16	Resistance	XXIV	Q	22	32	22	32	22	32	Ohm
Y	Threshold	XXV	X	See table XXV						
Z1-Z7	Disable time	XXVI	Q	5	46	5	46	5	46	nS
Z8-Z14	Disable time	XXVI	Q	5	90	5	90	5	90	nS
Z15-Z30	Disable time	XXVI	Q	5	46	5	46	5	46	nS
Z31-Z46	Disable time	XXVI	Q	5	90	5	90	5	90	nS

- See footnotes on next page

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/360-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 ± 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc ± 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

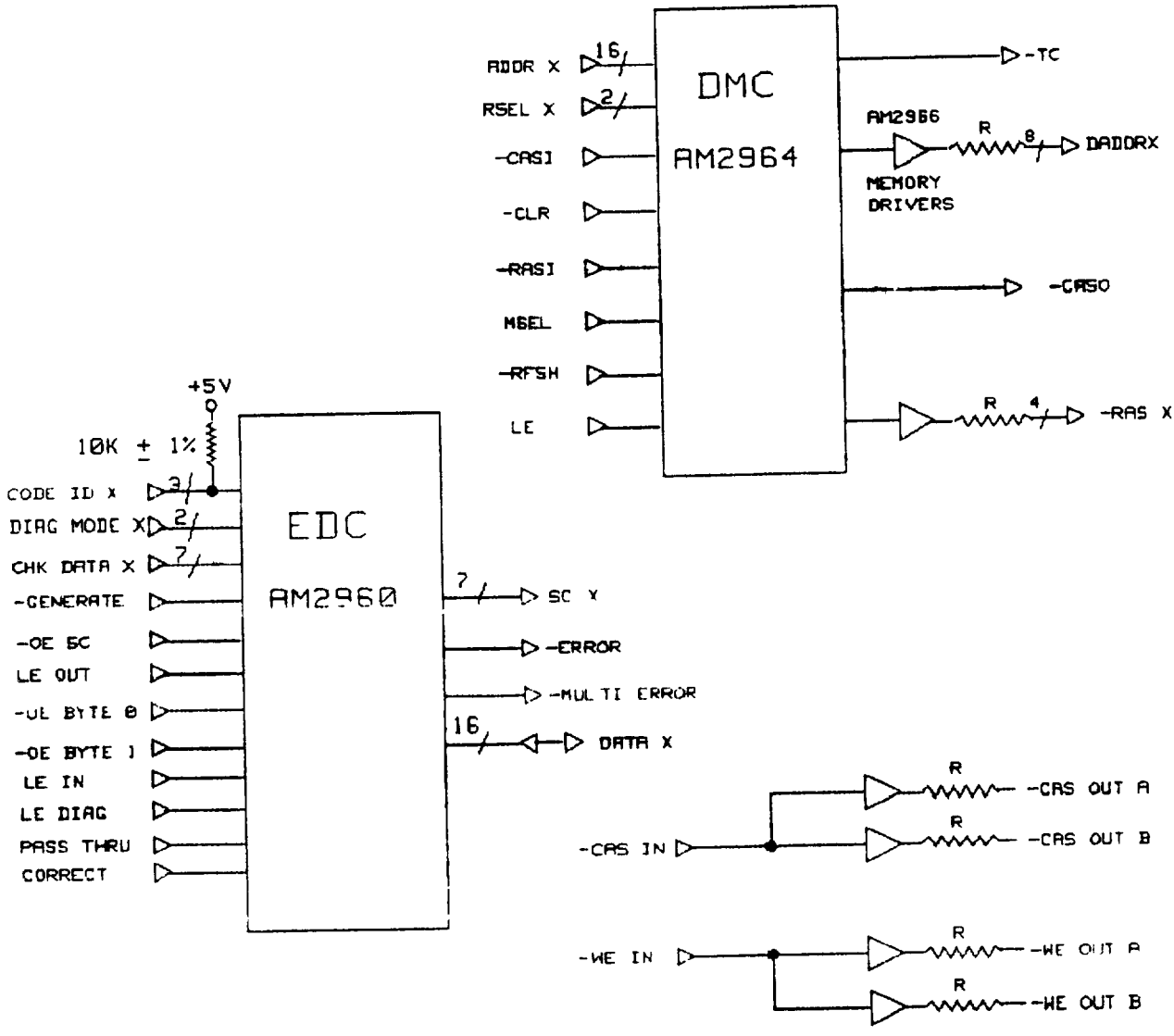
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTE:

1. R (series damping resistor) = 6.8 ohm ± 2 percent.

FIGURE 1. DFH functional block diagram.

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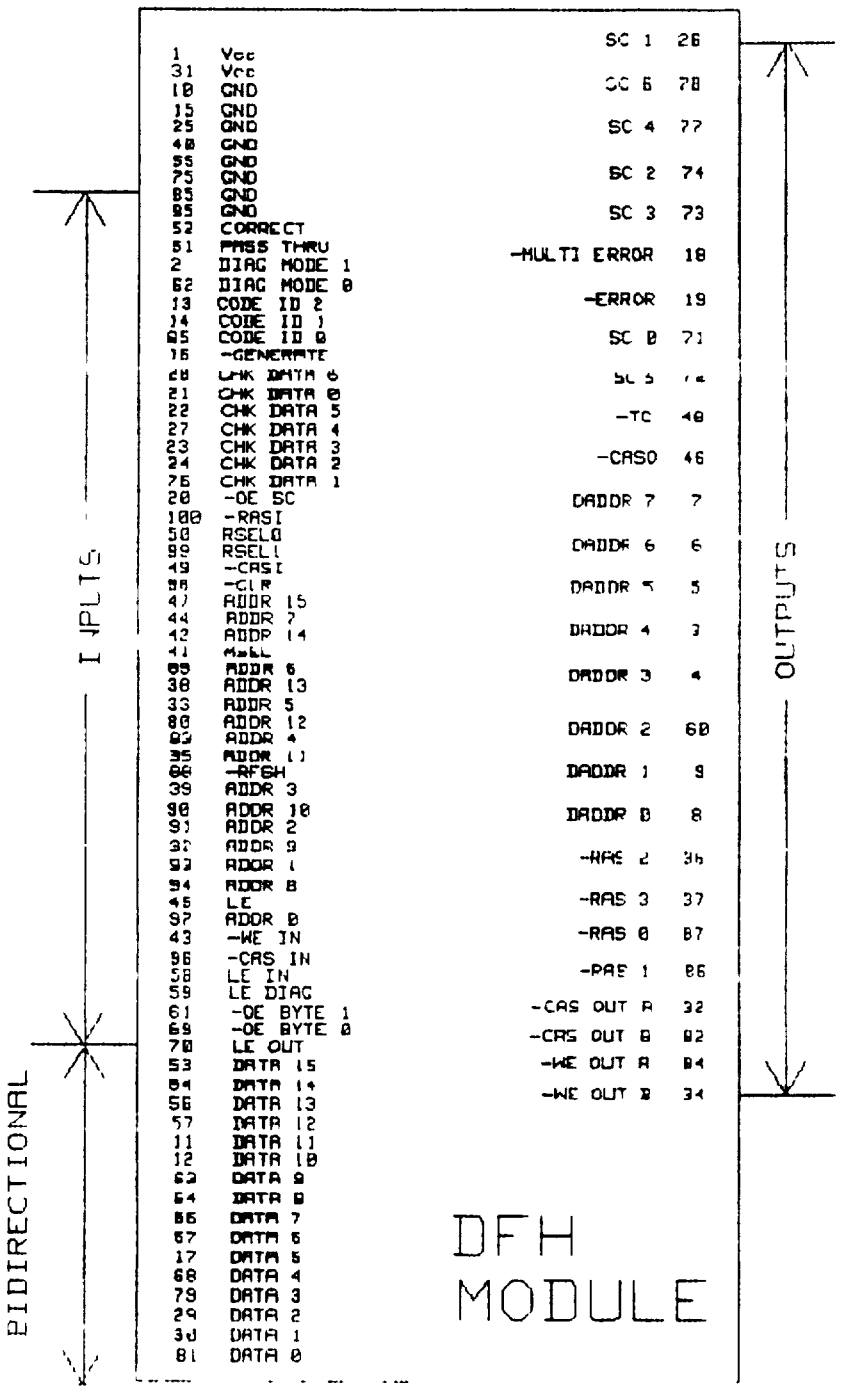


FIGURE 2. DFH pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Z shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E, or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

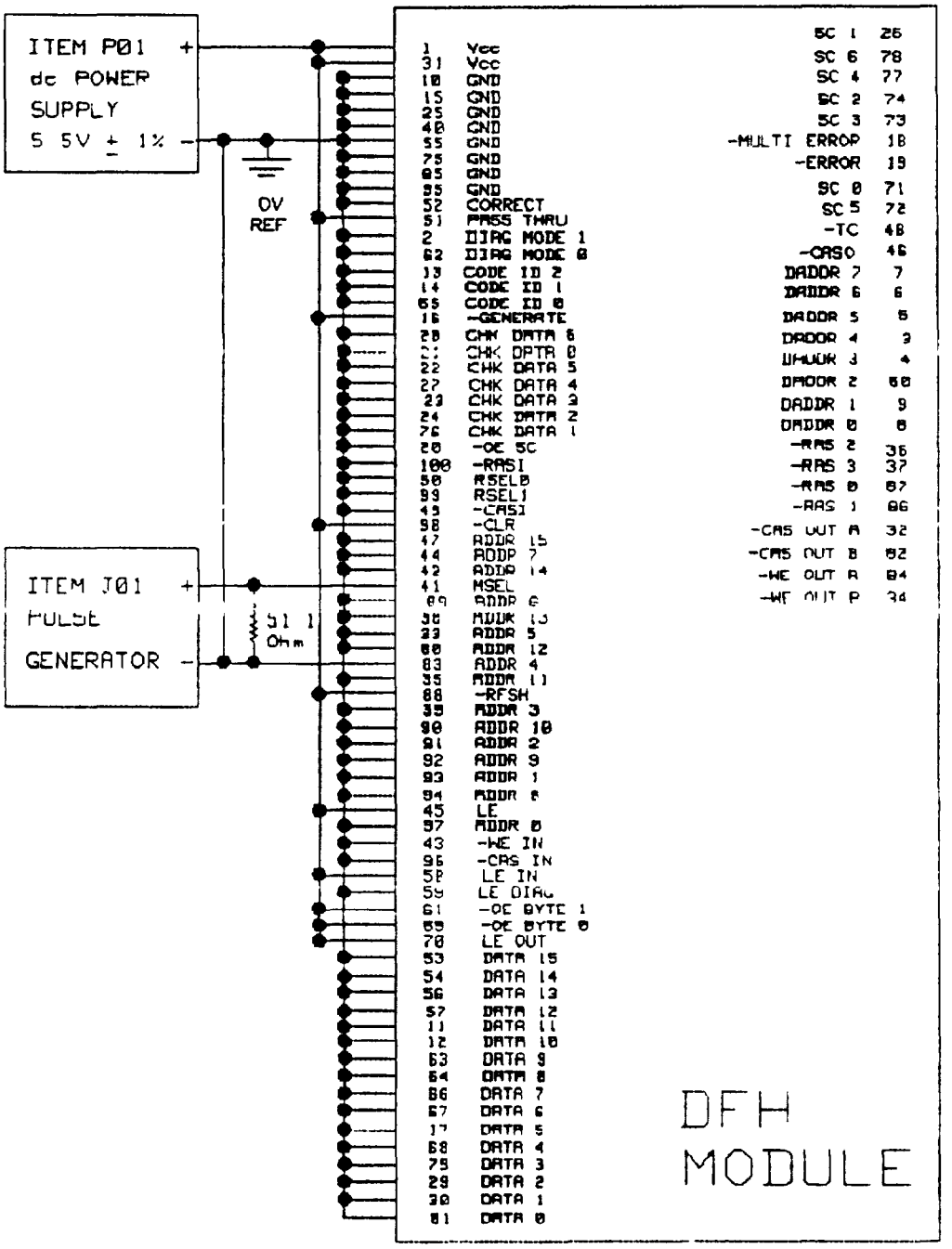
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Z in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through U, X, Y and Z shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3. The input waveforms are as indicated on figure 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Load each output as specified in the output voltage test.

FIGURE 3. DFH operating temperature and life test circuit.

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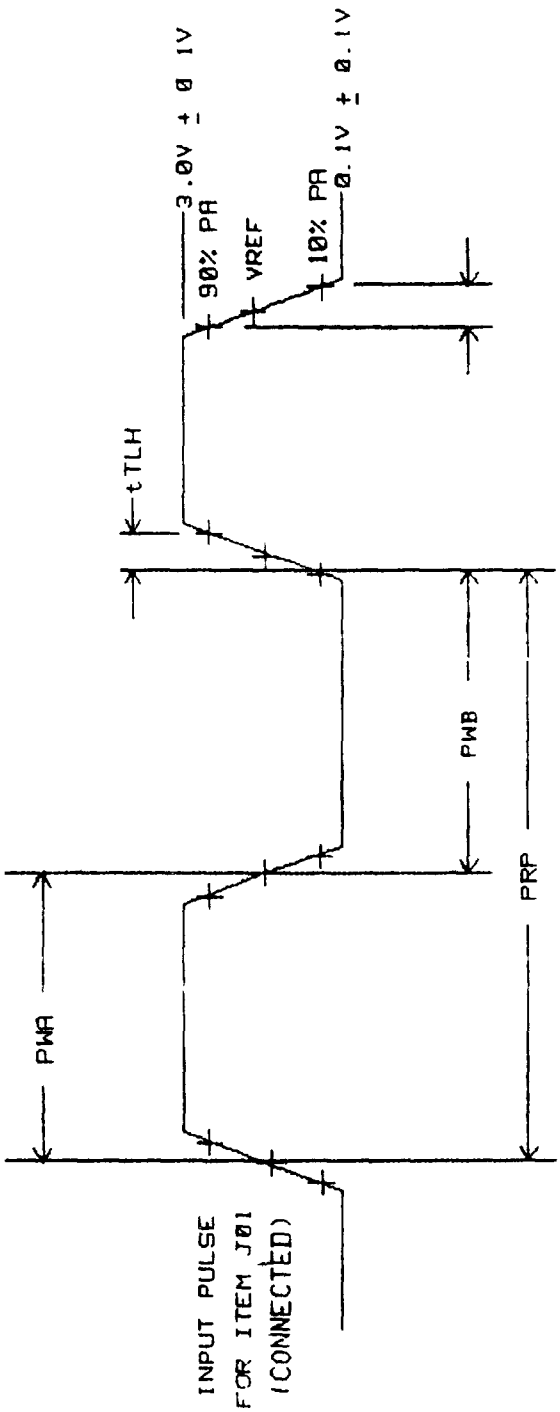


FIGURE 4. Operating temperature and life test waveforms.

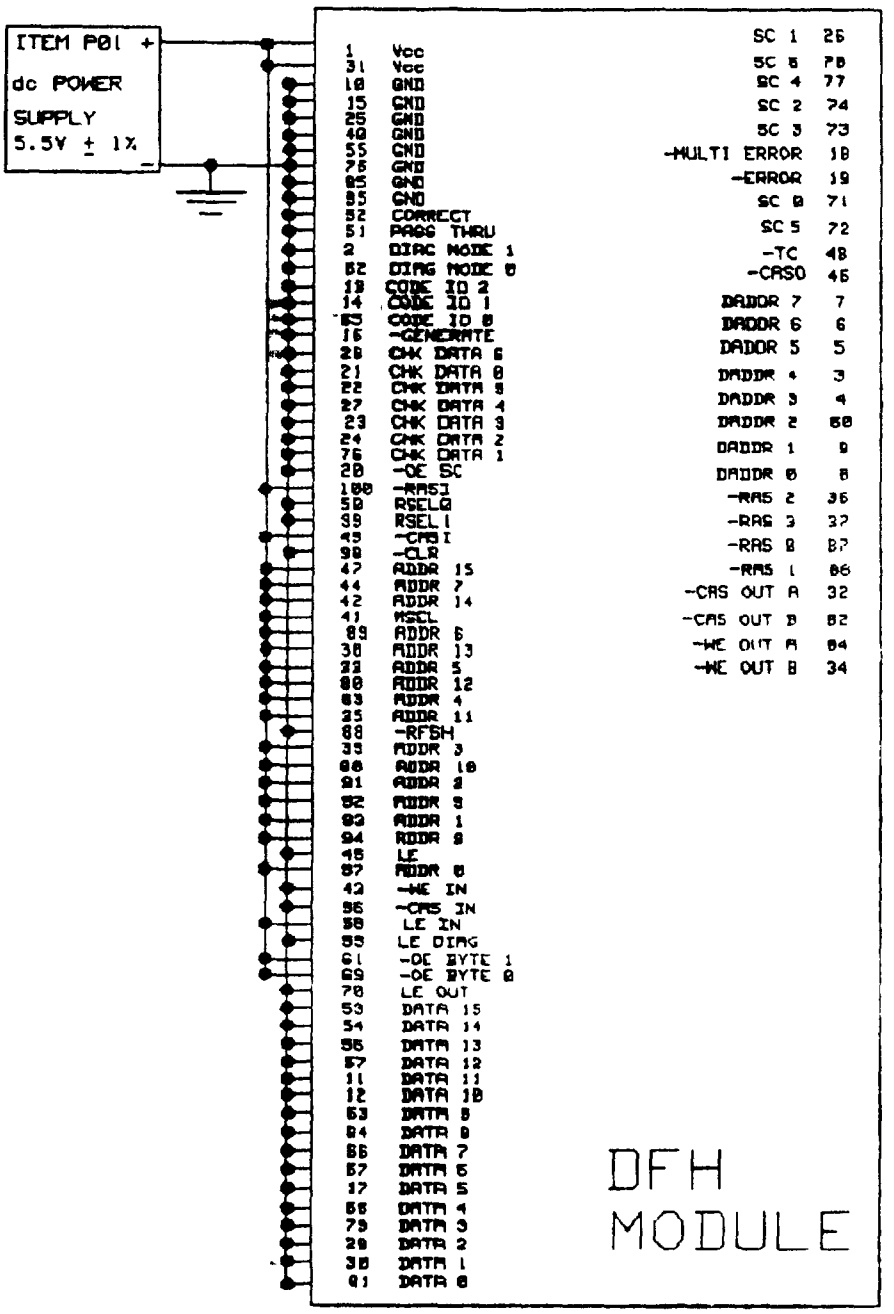
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NOTE:

1. Input pulse parameters for item J01:

   PWA = 250 ± 25 nS.
   PWB = 250 ± 25 nS.
   PRP = 500 ± 50 nS.
   tTLH = tTHL = 50 ± 10 nS.

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NOTE:

1. Load each output as specified in the output voltage test.

FIGURE 5. DFH component temperature test circuit.

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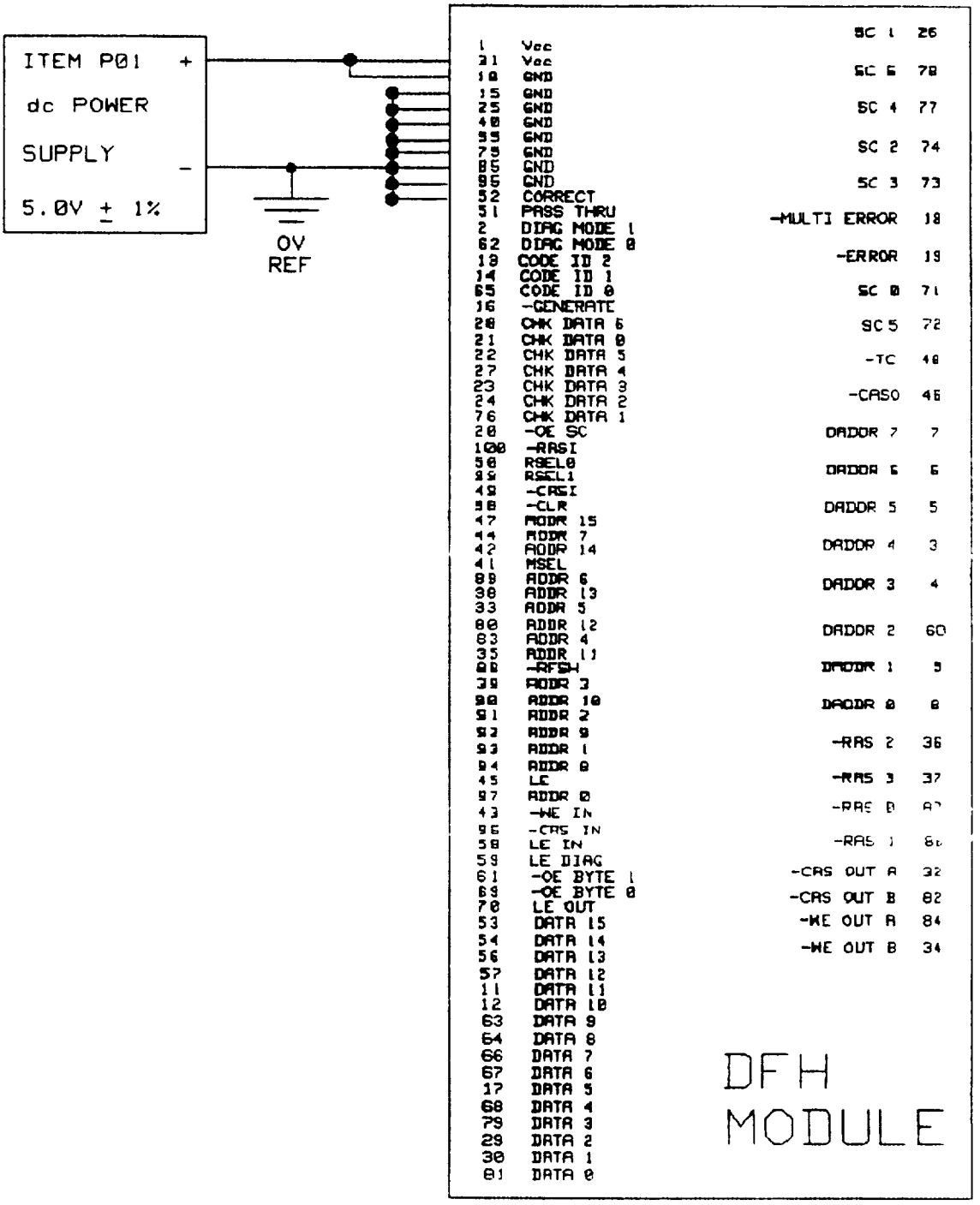


FIGURE 6. DFH vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through U, X, Y, and Z shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through U, X, Y and Z shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 4.547 W (max); 3.114 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DATA 0	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 1	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 2	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 3	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 4	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 5	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 6	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 7	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 8	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 9	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 10	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 11	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 12	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 13	-1.5	0.8	2.0	-0.41	70	1000	35

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
DATA 14	-1.5	0.8	2.0	-0.41	70	1000	35
DATA 15	-1.5	0.8	2.0	-0.41	70	1000	35
CODE ID 0	-1.5	0.8	2.0	-0.89	-0.30	1000	35
CODE ID 1	-1.5	0.8	2.0	-0.89	-0.30	1000	35
CODE ID 2	-1.5	0.8	2.0	-0.89	-0.30	1000	35
CHK DATA 0	-1.5	0.8	2.0	-0.36	50	1000	40
CHK DATA 1	-1.5	0.8	2.0	-0.36	50	1000	40
CHK DATA 2	-1.5	0.8	2.0	-0.36	50	1000	40
CHK DATA 3	-1.5	0.8	2.0	-0.36	50	1000	40
CHK DATA 4	-1.5	0.8	2.0	-0.36	50	1000	40
CHK DATA 5	-1.5	0.8	2.0	-0.36	50	1000	40
CHK DATA 6	-1.5	0.8	2.0	-0.36	50	1000	40
PASS THRU	-1.5	0.8	2.0	-0.36	50	1000	35
LE IN	-1.5	0.8	2.0	-0.36	50	1000	35

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
LE DIAG	-1.5	0.8	2.0	-0.36	50	1000	35
-OE BYTE 0	-1.5	0.8	2.0	-0.36	50	1000	35
-OE BYTE 1	-1.5	0.8	2.0	-0.36	50	1000	35
CORRECT	-1.5	0.8	2.0	-0.36	50	1000	35
LE OUT	-1.5	0.8	2.0	-0.36	50	1000	35
DIAG MODE 0	-1.5	0.8	2.0	-0.36	50	1000	35
DIAG MODE 1	-1.5	0.8	2.0	-0.36	50	1000	35
-GENERATE	-1.5	0.8	2.0	-0.36	50	1000	35
-OE SC	-1.5	0.8	2.0	-0.36	50	1000	35
LE	-1.5	0.8	2.0	-0.40	20	100	35
-CLR	-1.5	0.8	2.0	-0.40	20	100	35
RSEL0	-1.5	0.8	2.0	-0.40	20	100	35
RSEL1	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 0	-1.5	0.8	2.0	-0.40	20	100	35

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
ADDR 1	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 2	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 3	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 4	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 5	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 6	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 7	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 8	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 9	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 10	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 11	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 12	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 13	-1.5	0.8	2.0	-0.40	20	100	35
ADDR 14	-1.5	0.8	2.0	-0.40	20	100	35

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
ADDR 15	-1.5	0.8	2.0	-0.40	20	100	35
-WE IN	-1.2	0.8	2.0	-0.40	40	200	40
-CAS IN	-1.2	0.8	2.0	-0.40	40	200	40
MSEL	-1.5	0.8	2.0	1.6	50	1000	35
-RASI	-1.5	0.8	2.0	-3.2	100	2000	35
-RFSH	-1.5	0.8	2.0	-1.6	50	1000	35
-CASI	-1.5	0.8	2.0	-1.6	50	1000	35

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
DATA 0	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 1	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 2	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 3	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 4	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 5	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 6	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 7	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 8	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 9	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 10	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 11	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 12	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 13	2.4	0.5	-800	8.0	70	-410	-85	35

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
DATA 14	2.4	0.5	-800	8.0	70	-410	-85	35
DATA 15	2.4	0.5	-800	8.0	70	-410	-85	35
SC 0	2.4	0.5	-800	8.0	50	-50	-85	40
SC 1	2.4	0.5	-800	8.0	50	-50	-85	40
SC 2	2.4	0.5	-800	8.0	50	-50	-85	40
SC 3	2.4	0.5	-800	8.0	50	-50	-85	40
SC 4	2.4	0.5	-800	8.0	50	-50	-85	40
SC 5	2.4	0.5	-800	8.0	50	-50	-85	40
SC 6	2.4	0.5	-800	8.0	50	-50	-85	40
ERROR	2.4	0.5	-800	8.0	NA	NA	-85	NA
-MULTI ERROR	2.4	0.5	-800	8.0	NA	NA	-85	NA
DADDR 0	3.34	0.51	-1000	1.0	NA	NA	-200	NA
DADDR 1	3.34	0.51	-1000	1.0	NA	NA	-200	NA
DADDR 2	3.34	0.51	-1000	1.0	NA	NA	-200	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
DADDR 3	3.34	0.51	-1000	1.0	NA	NA	-200	NA
DADDR 4	3.34	0.51	-1000	1.0	NA	NA	-200	NA
DADDR 5	3.34	0.51	-1000	1.0	NA	NA	-200	NA
DADDR 6	3.34	0.51	-1000	1.0	NA	NA	-200	NA
DADDR 7	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-RAS 0	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-RAS 1	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-RAS 2	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-RAS 3	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-CAS OUT A	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-CAS OUT B	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-WE OUT A	3.34	0.51	-1000	1.0	NA	NA	-200	NA
-WE OUT B	3.34	0.51	-1000	1.0	NA	NA	-200	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (uA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios min (mA)	Cout max (pF)
-TC	2.5	0.5	-1000	8.0	NA	NA	-100	NA
-CASO	3.5	0.5	-1000	16.0	NA	NA	-100	NA

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test and the disable time test.

6.4 Module function. The module function is printed herein in the appendix.

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
DATA 0	81	I/O	These are the 16 bidirectional data lines to the Error Detection and Correction (EDC). They provide input to the data input latch and diagnostic latch. They also receive output from the data output latch. DATA 0 is the least significant bit and DATA 15 is the most significant bit.
DATA 1	30	I/O	
DATA 2	29	I/O	
DATA 3	79	I/O	
DATA 4	68	I/O	
DATA 5	17	I/O	
DATA 6	67	I/O	
DATA 7	66	I/O	
DATA 8	64	I/O	
DATA 9	63	I/O	
DATA 10	12	I/O	
DATA 11	11	I/O	
DATA 12	57	I/O	
DATA 13	55	I/O	
DATA 14	54	I/O	
DATA 15	53	I/O	
CODE ID 0	65	I	Code identification inputs. These three bits identify the size of the total data word to be processed, and which 16-bit slice of larger data words is being processed by the EDC.
CODE ID 1	14	I	
CODE ID 2	13	I	
CHK DATA 0	21	I	These are the seven check bit input lines to the EDC. The check bit lines are used to input bits for error detection. They are used to input syndrome bits for error correction in 32 and 64 bit configurations.
CHK DATA 1	76	I	
CHK DATA 2	24	I	
CHK DATA 3	23	I	
CHK DATA 4	27	I	
CHK DATA 5	22	I	
CHK DATA 6	28	I	

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
PASS THRU	51	I	Pass thru input. This line, when high, forces the contents of the EDC check bit input latch onto the syndrome/check bit outputs (SC 0-6) and the unmodified contents of the data input latch onto the inputs of the data output latch.
LE IN	58	I	Latch Enable - data input latch. Controls latching of the input data into the EDC. When high, the data input latch and check bit input latch follow the input data and input check bits. When low, the data input latch and check bit input latch are latched to their previous state.
LE DIAG	59	I	Latch Enable - diagnostic latch input. When high, the diagnostic latch follows the 16-bit data on the input lines. When low, the outputs of the diagnostic latch are latched to their previous states. The diagnostic latch holds diagnostic check bits, and internal control signals for CODE ID 0-2, DIAG MODE 0-1, CORRECT and PASS THRU.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-OE BYTE 0 -OE BYTE 1	69 61	I I	Output Enable - bytes 0 and 1, data output latch. These lines control the 3-state outputs for each of the two bytes of the EDC data output latch. When low, these lines enable the data output latch and, when high, these lines force the data output latch into the high impedance state. The two enable lines can be separately controlled to enable only one byte of the data output latch at a time.
CORRECT	52	I	Correct input. When high, this signal allows the correction network to correct any single-bit error in the EDC data input latch by complementing the bit-in-error before putting it onto the data output latch. When low, the EDC will drive data directly from the data input latch to the data output latch without correction.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
LE OUT	70	I	Latch Enable - data output latch. Controls the latching of the EDC data output latch. When low, the data output latch is latched to its previous state. When high, the data output latch follows the output of the data input latch as modified by the correction logic network. In correct mode, single-bit errors are corrected by the network before loading into the data output latch. In detect mode, the contents of the data input latch are passed through the correction network unchanged into the data output latch. The inputs to the data output latch are unspecified if the EDC is in generate mode.
DIAG MODE 0 DIAG MODE 1	62 2	I I	Diagnostic Mode Select. These two lines control the initialization and diagnostic operation of the EDC.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-GENERATE	16	I	Generate check bits, active low input. When this input is low, the EDC is in the check bit generation mode. When high, the EDC is in the detect mode or correct mode. In the generate mode, the circuit generates the check bits or partial check bits specific to the data in the data input latch. The generated check bits are placed on the SCX outputs. In the detect or correct modes, the EDC detects single and multiple errors, and generates syndrome bits based upon the contents of data input latch and check bit input latch. In correct mode, single-bit errors are also automatically corrected. Corrected data is placed at the inputs of the data output latch. The syndrome result is placed on the SCX outputs and indicates in a coded form the number of errors and the bit-in-error.
-OE SC	20	I	Output Enable - Syndrome/Checks bits. When low, the 3-state output lines SC 0 - SC 6 are enabled. When high, the SCX outputs are in the high impedance state.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-RFSH	88	I	The Dynamic Memory Controller (DMC) Refresh control input determines the outputs of the Row Address Strobe (RAS) and Column Address Strobe (CAS) buffers. When low, this input switches the address output multiplexer to output the inverted contents of the 8-bit refresh counter. A low on this input also inhibits the CAS buffer and changes the mode of the RAS decoder from one-of-four to four-of-four so that all four RAS decoder outputs (-RAS 0, -RAS 1, RAS 2, -RAS 3) go low in response to a low input at -RASI. This action refreshes one row address in each of the four RAS decoded memory banks. The refresh counter is advanced at the end of each refresh cycle by the low-to-high transition of -RFSH or -RASI (whichever occurs first). In burst mode refresh, -RFSH may be held low and refresh accomplished by toggling -RASI.
-CASI	49	I	This is the ADC Column Address Strobe Input. A low input at -CASI will result in an active low output at -CASO unless a refresh cycle is in progress (-RFSH = low).

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
MSEL	41	I	The ADC Multiplexer Select input determines whether low order or high order address inputs appear at the module outputs DADDR 0 - DADDR 7. When MSEL is high, the low order address inputs (signals ADDR 0 - ADDR 7) are connected to the outputs. When MSEL is low, the high order address inputs (signals ADDR 8 - ADDR 15) are connected to the outputs.
LE	45	I	LE is the ADC address latch enable input. A high input at LE causes the two 8-bit address latches and the 2-bit -RAS select input latch to go transparent, accepting new input data. A low input on LE latches the input data which meets setup and hold time specifications.
-CLR	98	I	-CLR is the ADC refresh counter clear input. A low input at -CLR resets the refresh counter to all zeroes which corresponds to refresh address output of all ones.
RSEL0 RSEL1	50 99	I I	These are the ADC -RAS decoder select inputs. Latched data at these inputs (normally higher order addresses) are decoded by the -RAS decoder to "-RAS select" one of four banks of memory with the -RAS 0, -RAS 1, -RAS 2 or -RAS 3 outputs.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
ADDR 0 ADDR 1 ADDR 2 ADDR 3 ADDR 4 ADDR 5 ADDR 6 ADDR 7	97 93 91 39 83 33 89 44	I I I I I I I I	These low order address inputs are used to input eight row address inputs into the ADC. These signal inputs drive the outputs (DADDR 0 - DADDR 7) when the signal MSEL is high.
ADDR 8 ADDR 9 ADDR 10 ADDR 11 ADDR 12 ADDR 13 ADDR 14 ADDR 15	94 92 90 35 80 38 42 47	I I I I I I I I	These high order address inputs are used to input eight column address inputs into the ADC. These signal inputs drive the outputs (DADDR 0 - DADDR 7) when the signal MSEL is low.
-WE IN	43	I	This signal is the active low write enable. Data is written into the Dynamic Random Access Memory (DRAM) array when -WE IN is low and read from the memory when it is high.
-CAS IN	96	I	This input signal is buffered by the memory driver.
-RASI	100	I	This is the ADC Row Address Strobe Input. During normal memory cycles, the selected -RAS decoder output (-RAS 0 to -RAS 3) will go low in response to a low input at -RASI. During refresh (-RFSH = low), all -RAS outputs go low in response to -RASI going low.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
SC 0 SC 1 SC 2 SC 3 SC 4 SC 5 SC 6	71 26 74 73 77 72 78	O O O O O O O	Syndrome/Check bit outputs. These seven lines hold the check/partial-check bits when the EDC is in generate mode, and will hold the syndrome/partial syndrome bits when the device is in detect or correct modes. These are 3-state outputs.
-ERROR	19	O	Error detected output. When the EDC is in detect or correct mode, this output will go low if one or more syndrome bits are asserted, meaning there are one or more bit errors in the data or check bits. If no syndrome bits are asserted, there are no errors detected and the output will be high. In generate mode, -ERROR is forced high. (In a 64-bit configuration, -ERROR must be externally implemented.)
-MULTI ERROR	18	O	Multiple errors detected output. When the EDC is in detect or correct mode, this output, if low, indicates that there are two or more bit errors that have been detected. If high, this indicates that either one or no errors have been detected. In generate mode, -MULTI ERROR is forced high. (In a 64-bit configuration -MULTI ERROR must be externally implemented.)

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-RAS 0 -RAS 1 -RAS 2 -RAS 3	87 86 36 37	O O O O	These ADC signals are the Row Address Strobe (-RASI) outputs. Each signal provides a row address strobe for one of the four banks of memory. Each will go low only when selected by RSEL0 and RSEL1 and only when -RASI goes low. All -RAS 0 through -RAS 3 outputs go low in response to -RASI when -RFSH goes low. These outputs are buffered and contain $\geq$ 6.8 Ohm resistor in series to limit undershoot at the DRAM inputs.
DADDR 0 DADDR 1 DADDR 2 DADDR 3 DADDR 4 DADDR 5 DADDR 6 DADDR 7	8 9 60 4 3 5 6 7	O O O O O O O O	These signal lines are the Random Access Memory (RAM) address outputs of the ADC buffered by a memory driver with a series 6.8 ohm resistor to limit undershoot at the DRAM inputs. The eight bit width is designed to directly address up to 64K of memory.
-CAS OUT A -CAS OUT B	32 82	O O	These active low signals are the outputs of the -CAS IN input. Both are buffered and contain a 6.8 ohm resistor in series to limit undershoot at the DRAM inputs.
-WE OUT A -WE OUT B	84 34	O O	These signals are the noninverted buffered outputs of the -WE IN input. They are connected to the DRAM array and contain a 6.8 ohm resistor in series to limit undershoot at the DRAM inputs.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
-TC	48	O	The ADC Terminal Count output indicates that the refresh counter has been sequenced through 256 refresh address. This output goes low indicating the terminal count. The -TC output remains low until the refresh counter is advanced by the rising edge of -RASI or -RFSH.
-CASO	46	O	This is the ADC Column Address Strobe output directly from the AM2964. The active low output at -CASO strobes the column address into the dynamic RAM. -CASO is inhibited during refresh (-RFSH = low).

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6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 7.

(j) Vol test - table XI, figure 7.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 8.

(o) Dynamic test - tables XVI and XVII, figures 9 and 10.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figure 11.

(r) Output crosstalk tests - tables XX and XXI, figures 12 and 13.

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(s) C1/O test - table XXII.

(t) Cout test - table XXIII.

(u) Resistance test - table XXIV, figure 14.

(v) Threshold test - table XXV.

(w) Disable time test - table XXVI, figures 15 and 16.

6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
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(Project 5963-N745)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A A

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40				
B5	Continuity, GND	40	55				
B6	Continuity, GND	55	75				
B7	Continuity, GND	75	85				
B8	Continuity, GND	85	95	0.0	1.0	Ohm	1-4
B9	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B10	Continuity, GND	10	15				
B11	Continuity, GND	15	25				
B12	Continuity, GND	25	40				
B13	Continuity, GND	40	55				
B14	Continuity, GND	55	75				
B15	Continuity, GND	75	85				
B16	Continuity, GND	85	95	0.0	0.1	Ohm	1-4

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NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B8 B9-B16

2. The ohmmeter used shall be item R10.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15, 25,40, 55,75, 85,95	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	1.24 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

6. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	DATA 0	81	5.5	0.0	1.0	mA	1-4
D2	DATA 1	30					
D3	DATA 2	29					
D4	DATA 3	79					
D5	DATA 4	68					
D6	DATA 5	17					
D7	DATA 6	67					
D8	DATA 7	66					
D9	DATA 8	64					
D10	DATA 9	63					
D11	DATA 10	12					
D12	DATA 11	11					
D13	DATA 12	57					
D14	DATA 13	56					
D15	DATA 14	54					
D16	DATA 15	53					
D17	CODE ID 0	65					
D18	CODE ID 1	14					
D19	CODE ID 2	13					
D20	CHK DATA 0	21	5.5	0.0	1.0	mA	1-4

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	CHK DATA 1	76	5.5	0.0	1.0	mA	1-4
D22	CHK DATA 2	24					
D23	CHK DATA 3	23					
D24	CHK DATA 4	27					
D25	CHK DATA 5	22					
D26	CHK DATA 6	28					
D27	PASS THRU	51					
D28	LE IN	58					
D29	LE DIAG	59					
D30	-OE BYTE 0	69					
D31	-OE BYTE 1	61					
D32	CORRECT	52					
D33	LE OUT	70					
D34	DIAG MODE 0	62					
D35	DIAG MODE 1	2					
D36	-GENERATE	16					
D37	-OE SC	20					
D38	-RFSH	88					
D39	-CASI	49					
D40	MSEL	41	5.5	0.0	1.0	mA	1-4

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	LE	45	5.5	0.0	0.1	mA	1-4
D42	-CLR	98					
D43	RSEL0	50					
D44	RSEL1	99					
D45	ADDR 0	97					
D46	ADDR 1	93					
D47	ADDR 2	91					
D48	ADDR 3	39					
D49	ADDR 4	83					
D50	ADDR 5	33					
D51	ADDR 6	89					
D52	ADDR 7	44					
D53	ADDR 8	94					
D54	ADDR 9	92					
D55	ADDR 10	90					
D56	ADDR 11	35					
D57	ADDR 12	80					
D58	ADDR 13	38					
D59	ADDR 14	42					
D60	ADDR 15	47	5.5	0.0	0.1	mA	1-4

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D61	-WE IN	43	7.0	0.0	0.2	mA	1-4
D62	-CAS IN	96	7.0	0.0	0.2	mA	1-4
D63	-RASI	100	5.5	0.0	2.0	mA	1-4

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D16, D20-D63
	5.5 V	+ 0mV,-55mV	D17-D19
Power supply current limit	1.24 A	NA	D1-D63
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D16, D20-60,D63
	5.5 V	+ 0mV,-55mV	D17-D19
	5.5 V	+ 0mV,-70mV	D61,D62
Pattern voltage high, (VH)	3.0 V	± 100 mV	D1-D63
low, (VL)	0.0 V	± 100 mV	D1-D63
Input current measurement	NA	± 50 uA	D1-D40
	NA	± 5 uA	D41-D60
	NA	± 10 uA	D61,D62
	NA	± 100 uA	D63

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	DATA 0	81	-18.0	-1.5	-0.2	V	1-4
E2	DATA 1	30					
E3	DATA 2	29					
E4	DATA 3	79					
E5	DATA 4	68					
E6	DATA 5	17					
E7	DATA 6	67					
E8	DATA 7	66					
E9	DATA 8	64					
E10	DATA 9	63					
E11	DATA 10	12					
E12	DATA 11	11					
E13	DATA 12	57					
E14	DATA 13	56					
E15	DATA 14	54					
E16	DATA 15	53					
E17	CODE ID 0	65					
E18	CODE ID 1	14					
E19	CODE ID 2	13					
E20	CHK DATA 0	21	-18.0	-1.5	-0.2	V	1-4

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	CHK DATA 1	76	-18.0	-1.5	-0.2	V	1-4
E22	CHK DATA 2	24					
E23	CHK DATA 3	23					
E24	CHK DATA 4	27					
E25	CHK DATA 5	22					
E26	CHK DATA 6	28					
E27	PASS THRU	51					
E28	LE IN	58					
E29	LE DIAG	59					
E30	-OE BYTE 0	69					
E31	-OE BYTE 1	61					
E32	CORRECT	52					
E33	LE OUT	70					
E34	DIAG MODE 0	62					
E35	DIAG MODE 1	2					
E36	-GENERATE	16					
E37	-OE SC	20	-18.0	-1.5	-0.2	V	1-4
E38	-RFSH	88	-18.0	-1.5	-0.4	V	1-4
E39	-CASI	49					
E40	MSEL	41	-18.0	-1.5	-0.4	V	1-4

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
E41	LE	45	-18.0	-1.5	-0.4	V	1-4
E42	-CLR	98					
E43	RSEL0	50					
E44	RSEL1	99					
E45	ADDR 0	97					
E46	ADDR 1	93					
E47	ADDR 2	91					
E48	ADDR 3	39					
E49	ADDR 4	83					
E50	ADDR 5	33					
E51	ADDR 6	89					
E52	ADDR 7	44					
E53	ADDR 8	94					
E54	ADDR 9	92					
E55	ADDR 10	90					
E56	ADDR 11	35					
E57	ADDR 12	80					
E58	ADDR 13	38					
E59	ADDR 14	42	-18.0	-1.5	-0.4	V	1-4

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E60	ADDR 15	47	-18.0	-1.5	-0.4	V	1-4
E61	-RASI	100	-18.0	-1.5	-0.4	V	1-4
E62	-WE IN	43	-18.0	-1.2	-0.4	V	1-4
E63	-CAS IN	96	-18.0	-1.2	-0.4	V	1-4

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	E1-E63
Power supply current limit	1.24 A	NA	E1-E63
Forcing current, (Iin)	-18 mA	± 1.8 mA	E1-E63
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E63 E1-E63
Input voltage measurement	NA NA NA	± 65 mV ± 55 mV ± 40 mV	E1-E37 E38-E61 E62,E63

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	DATA 0	81	0.0	-85	-25	mA	1-5
F2	DATA 1	30					
F3	DATA 2	29					
F4	DATA 3	79					
F5	DATA 4	68					
F6	DATA 5	17					
F7	DATA 6	67					
F8	DATA 7	66					
F9	DATA 8	64					
F10	DATA 9	63					
F11	DATA 10	12					
F12	DATA 11	11					
F13	DATA 12	57					
F14	DATA 13	56					
F15	DATA 14	54					
F16	DATA 15	53					
F17	SC 0	71					
F18	SC 1	26					
F19	SC 2	74					
F20	SC 3	73	0.0	-85	-25	mA	1-5

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	SC 4	77	0.0	-85	-25	mA	1-5
F22	SC 5	72					
F23	SC 6	78					
F24	-ERROR	19					
F25	-MULTI ERROR	18	0.0	-85	-25	mA	1-5
F26	-RAS 0	87	0.0	-200	-60	mA	1-5
F27	-RAS 1	86					
F28	-RAS 2	36					
F29	-RAS 3	37					
F30	DADDR 0	8					
F31	DADDR 1	9					
F32	DADDR 2	60					
F33	DADDR 3	4					
F34	DADDR 4	3					
F35	DADDR 5	5					
F36	DADDR 6	6					
F37	DADDR 7	7					
F38	-CAS OUT A	32					
F39	-CAS OUT B	82					
F40	-WE OUT A	84	0.0	-200	-60	mA	1-5

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	-WE OUT B	34	0.0	-200	-60	mA	1-5
F42	-TC	48	0.0	-100	-40	mA	1-5
F43	-CASO	46	0.0	-100	-40	mA	1-5

NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F43
Power supply current limit	1.24 A	NA	F1-F43
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F43
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	F1-F43 F1-F43
Output current measurement	NA NA NA	± 3 mA ± 7 mA ± 3 mA	F1-F25 F26-F41 F42,F43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Floating inputs may cause damage to the module. All inputs should be tied high (3.0 V) or low (0.0 V) during testing.

5. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	DATA 0	81	2.7	-70	70	uA	1-4
G2	DATA 1	30					
G3	DATA 2	29					
G4	DATA 3	79					
G5	DATA 4	68					
G6	DATA 5	17					
G7	DATA 6	67					
G8	DATA 7	66					
G9	DATA 8	64					
G10	DATA 9	63					
G11	DATA 10	12					
G12	DATA 11	11					
G13	DATA 12	57					
G14	DATA 13	56					
G15	DATA 14	54					
G16	DATA 15	53	2.7	-70	70	uA	1-4
G17	CODE ID 0	65	2.7	-295	-217	uA	1-4
G18	CODE ID 1	14					
G19	CODE ID 2	13	2.7	-295	-217	uA	1-4
G20	CHK DATA 0	21	2.7	0	50	uA	1-4

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	CHK DATA 1	76	2.7	0	50	uA	1-4
G22	CHK DATA 2	24					
G23	CHK DATA 3	23					
G24	CHK DATA 4	27					
G25	CHK DATA 5	22					
G26	CHK DATA 6	28					
G27	PASS THRU	51					
G28	LE IN	58					
G29	LE DIAG	59					
G30	-OE BYTE 0	69					
G31	-OE BYTE 1	61					
G32	CORRECT	52					
G33	LE OUT	70					
G34	DIAG MODE 0	62					
G35	DIAG MODE 1	2					
G36	-GENERATE	16					
G37	-OE SC	20					
G38	MSEL	41					
G39	-RFSH	88					
G40	-CASI	49	2.7	0	50	uA	1-4

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G41	ADDR 0	97	2.7	0	20	uA	1-4
G42	ADDR 1	93					
G43	ADDR 2	91					
G44	ADDR 3	39					
G45	ADDR 4	83					
G46	ADDR 5	33					
G47	ADDR 6	89					
G48	ADDR 7	44					
G49	ADDR 8	94					
G50	ADDR 9	92					
G51	ADDR 10	90					
G52	ADDR 11	35					
G53	ADDR 12	80					
G54	ADDR 13	38					
G55	ADDR 14	42					
G56	ADDR 15	47					
G57	RSEL0	50					
G58	RSEL1	99					
G59	-CLR	98					
G60	LE	45	2.7	0	20	uA	1-4

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G61	-WE IN	43	2.7	0	40	uA	1-4
G62	-CAS IN	96	2.7	0	40	uA	1-4
G63	-RASI	100	2.7	0	100	uA	1-4

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G63
Power supply current limit	1.24 A	NA	G1-G63
Forcing voltage, (Vih)	2.7 V	± 135 mV	G1-G63
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G63
low, (VL)	0.0 V	± 100 mV	G1-G63
Input current measurement	NA	± 7.0 uA	G1-G16
	NA	± 3.9 uA	G17-G19
	NA	± 2.5 uA	G20-G40
	NA	± 1.0 uA	G41-G60
	NA	± 2.0 uA	G61,G62
	NA	± 5.0 uA	G63

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Floating inputs may cause damage to the module. All inputs should be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	DATA 0	81	0.5	-410.0	0.0	uA	1-4
H2	DATA 1	30					
H3	DATA 2	29					
H4	DATA 3	79					
H5	DATA 4	68					
H6	DATA 5	17					
H7	DATA 6	67					
H8	DATA 7	66					
H9	DATA 8	64					
H10	DATA 9	63					
H11	DATA 10	12					
H12	DATA 11	11					
H13	DATA 12	57					
H14	DATA 13	56					
H15	DATA 14	54					
H16	DATA 15	53	0.5	-410.0	0.0	uA	1-4
H17	CHK DATA 0	21	0.5	-360.0	0.0	uA	1-4
H18	CHK DATA 1	76					
H19	CHK DATA 2	24					
H20	CHK DATA 3	23	0.5	-360.0	0.0	uA	1-4

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	CHK DATA 4	27	0.5	-360.0	0.0	uA	1-4
H22	CHK DATA 5	22					
H23	CHK DATA 6	28					
H24	PASS THRU	51					
H25	LE IN	58					
H26	-GENERATE	16					
H27	-OE SC	20					
H28	LE DIAG	59					
H29	CORRECT	52					
H30	-OE BYTE 0	69					
H31	-OE BYTE 1	61					
H32	DIAG MODE 0	62					
H33	DIAG MODE 1	2					
H34	LE OUT	70	0.5	-360.0	0.0	uA	1-4
H35	CODE ID 0	65	0.5	-886.0	-476.0	uA	1-4
H36	CODE ID 1	14					
H37	CODE ID 2	13	0.5	-886.0	-476.0	uA	1-4
H38	ADDR 0	97	0.4	-400.0	0.0	uA	1-4
H39	ADDR 1	93					
H40	ADDR 2	91	0.4	-400.0	0.0	uA	1-4

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	ADDR 3	39	0.4	-400.0	0.0	uA	1-4
H42	ADDR 4	83					
H43	ADDR 5	33					
H44	ADDR 6	89					
H45	ADDR 7	44					
H46	ADDR 8	94					
H47	ADDR 9	92					
H48	ADDR 10	90					
H49	ADDR 11	35					
H50	ADDR 12	80					
H51	ADDR 13	38					
H52	ADDR 14	42					
H53	ADDR 15	47					
H54	RSEL0	50					
H55	RSEL1	99					
H56	-CLR	98					
H57	LE	45					
H58	-WE IN	43					
H59	-CAS IN	96	0.4	-400.0	0.0	uA	1-4
H60	-RASI	100	0.4	-3.2	0.0	mA	1-4

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H61	-CASI	49	0.4	-1.6	0.0	mA	1-4
H62	MSEL	41					
H63	-RFSH	88	0.4	-1.6	0.0	mA	1-4

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H63
Power supply current limit	1.24 A	NA	H1-H63
Forcing voltage, (Vil)	0.5 V 0.4 V	± 25 mV ± 20 mV	H1-H37 H38-H63
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H63 H1-H63
Input current measurement	NA NA NA NA NA NA	± 20.5 uA ± 19.8 uA ± 20.5 uA ± 20.0 uA ± 160.0 uA ± 80.0 uA	H1-H16 H17-H34 H35-H37 H38-H59 H60 H61-H63

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Floating inputs may cause damage to the module. All inputs should be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	DATA 0	81	-0.8	2.4	4.5	V	1-5
J2	DATA 1	30					
J3	DATA 2	29					
J4	DATA 3	79					
J5	DATA 4	68					
J6	DATA 5	17					
J7	DATA 6	67					
J8	DATA 7	66					
J9	DATA 8	64					
J10	DATA 9	63					
J11	DATA 10	12					
J12	DATA 11	11					
J13	DATA 12	57					
J14	DATA 13	56					
J15	DATA 14	54					
J16	DATA 15	53					
J17	SC 0	71					
J18	SC 1	26					
J19	SC 2	74					
J20	SC 3	73	-0.8	2.4	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	SC 4	77	-0.8	2.4	4.5	V	1-5
J22	SC 5	72					
J23	SC 6	78					
J24	-ERROR	19					
J25	-MULTI ERROR	18	-0.8	2.4	4.5	V	1-5
J26	-CASO	46	-1.0	3.0	4.5	V	1-5
J27	-TC	48	-1.0	2.5	4.5	V	1-5
J28	DADDR 0	8	-1.0	3.34	4.5	V	1-5
J29	DADDR 1	9					
J30	DADDR 2	60					
J31	DADDR 3	4					
J32	DADDR 4	3					
J33	DADDR 5	5					
J34	DADDR 6	6					
J35	DADDR 7	7					
J36	-RAS 0	87					
J37	-RAS 1	86					
J38	-RAS 2	36					
J39	-RAS 3	37					
J40	-CAS OUT A	32	-1.0	3.34	4.5	V	1-5

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	-CAS OUT B	82	-1.0	3.34	4.5	V	1-5
J42	-WE OUT A	84	″	″	″	″	″
J43	-WE OUT B	34	-1.0	3.34	4.5	V	1-5

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J43
Power supply current limit	1.24 A	NA	J1-J43
Forcing current, (Ioh)	-0.8 mA -1.0 mA	± 8 uA ± 10 uA	J1-J25 J26-J43
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J43 J1-J43
Output voltage measurement	NA NA NA NA	± 105 mV ± 75 mV ± 100 mV ± 58 mV	J1-J25 J26 J27 J28-J43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

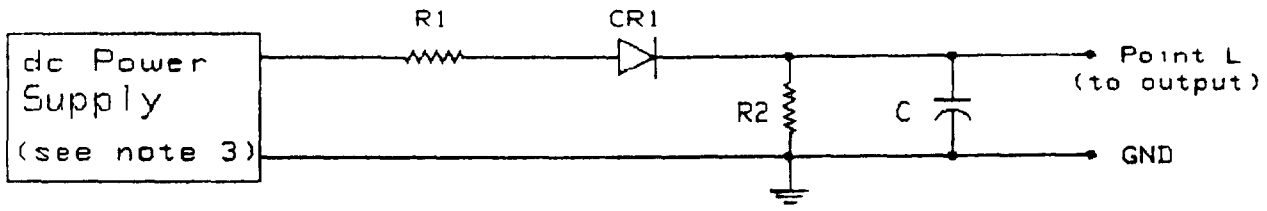
(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 7.

4. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

5. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 7. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply voltage set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are specified below.

5. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
3,4,5,6,7,8,9,32, 34,36,37,60,82, 84,86,87	3340	1.0	0.5
11,12,17,18,19, 26,29,30,53,54, 56,57,63,64,66, 67,68,71,72,73, 74,77,78,79,81	3000	8.0	0.5
46	3000	16.0	0.5
48	2500	8.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	DATA 0	81	8.0	0.0	0.5	V	1-5
K2	DATA 1	30					
K3	DATA 2	29					
K4	DATA 3	79					
K5	DATA 4	68					
K6	DATA 5	17					
K7	DATA 6	67					
K8	DATA 7	66					
K9	DATA 8	64					
K10	DATA 9	63					
K11	DATA 10	12					
K12	DATA 11	11					
K13	DATA 12	57					
K14	DATA 13	56					
K15	DATA 14	54					
K16	DATA 15	53					
K17	SC 0	71					
K18	SC 1	26					
K19	SC 2	74					
K20	SC 3	73	8.0	0.0	0.5	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K21	SC 4	77	8.0	0.0	0.5	V	1-5
K22	SC 5	72					
K23	SC 6	78					
K24	-ERROR	19					
K25	-MULTI ERROR	18					
K26	-TC	48	8.0	0.0	0.5	V	1-5
K27	-CASO	46	16.0	0.0	0.5	V	1-5
K28	DADDR 0	8	1.0	0.0	0.507	V	1-5
K29	DADDR 1	9					
K30	DADDR 2	60					
K31	DADDR 3	4					
K32	DADDR 4	3					
K33	DADDR 5	5					
K34	DADDR 6	6					
K35	DADDR 7	7					
K36	-RAS 0	87					
K37	-RAS 1	86					
K38	-RAS 2	36					
K39	-RAS 3	37					
K40	-CAS OUT A	32	1.0	0.0	0.507	V	1-5

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K41	-CAS OUT B	82	1.0	0.0	0.507	V	1-5
K42	-WE OUT A	84	⎮	⎮	⎮	⎮	⎮
K43	-WE OUT B	34	1.0	0.0	0.507	V	1-5

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K43
Power supply current limit	1.24 A	NA	K1-K43
Forcing current, (Iol)	8.0 mA 16.0 mA 1.0 mA	± 80 uA ± 160 uA ± 10 uA	K1-K26 K27 K28-K43
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K1-K43 K1-K43
Output voltage measurement	NA NA	± 25.0 mV ± 25.4 mV	K1-K27 K28-K43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. The load shall be as indicated on figure 7.

4. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

5. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
L1	DATA 0	81	2.4	-70	70	uA	1-4
L2	DATA 1	30					
L3	DATA 2	29					
L4	DATA 3	79					
L5	DATA 4	68					
L6	DATA 5	17					
L7	DATA 6	67					
L8	DATA 7	66					
L9	DATA 8	64					
L10	DATA 9	63					
L11	DATA 10	12					
L12	DATA 11	11					
L13	DATA 12	57					
L14	DATA 13	56					
L15	DATA 14	54					
L16	DATA 15	53	2.4	-70	70	uA	1-4
L17	SC 0	71	2.4	-50	50	uA	1-4
L18	SC 1	26					
L19	SC 2	74					
L20	SC 3	73	2.4	-50	50	uA	1-4

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	SC 4	77	2.4	-50	50	uA	1-4
L22	SC 5	72					
L23	SC 6	78	2.4	-50	50	uA	1-4

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L23
Power supply current limit	1.24 A	NA	L1-L23
Forcing voltage, (Voh)	2.4 V	± 120 mV	L1-L23
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L23 L1-L23
Output current measurement	NA NA	± 7 uA ± 5 uA	L1-L16 L17-L23

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	DATA 0	81	0.5	-410	0	uA	1-4
M2	DATA 1	30					
M3	DATA 2	29					
M4	DATA 3	79					
M5	DATA 4	68					
M6	DATA 5	17					
M7	DATA 6	67					
M8	DATA 7	66					
M9	DATA 8	64					
M10	DATA 9	63					
M11	DATA 10	12					
M12	DATA 11	11					
M13	DATA 12	57					
M14	DATA 13	56					
M15	DATA 14	54					
M16	DATA 15	53	0.5	-410	0	uA	1-4
M17	SC 0	71	0.5	-50	50	uA	1-4
M18	SC 1	26					
M19	SC 2	74					
M20	SC 3	73	0.5	-50	50	uA	1-4

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
M21	SC 4	77	0.5	-50	50	uA	1-4
M22	SC 5	72	|	|	|	|	|
M23	SC 6	78	0.5	-50	50	uA	1-4

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M23
Power supply current limit	1.24 A	NA	M1-M23
Forcing voltage, (Vol)	0.5 V	± 25 mV	M1-M23
Pattern voltage high, (VH)	3.0 V	± 100 mV	M1-M23
low, (VL)	0.0 V	± 100 mV	M1-M23
Output current measurement	NA	± 20.5 uA	M1-M16
	NA	± 5.0 uA	M17-M23

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31	41	827	mA	1-4

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.24 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 39.3 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-4
P2	Functional test maximum power supply voltage maximum pattern rate	1-4

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.24 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 150 pF (see figure 7). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

3. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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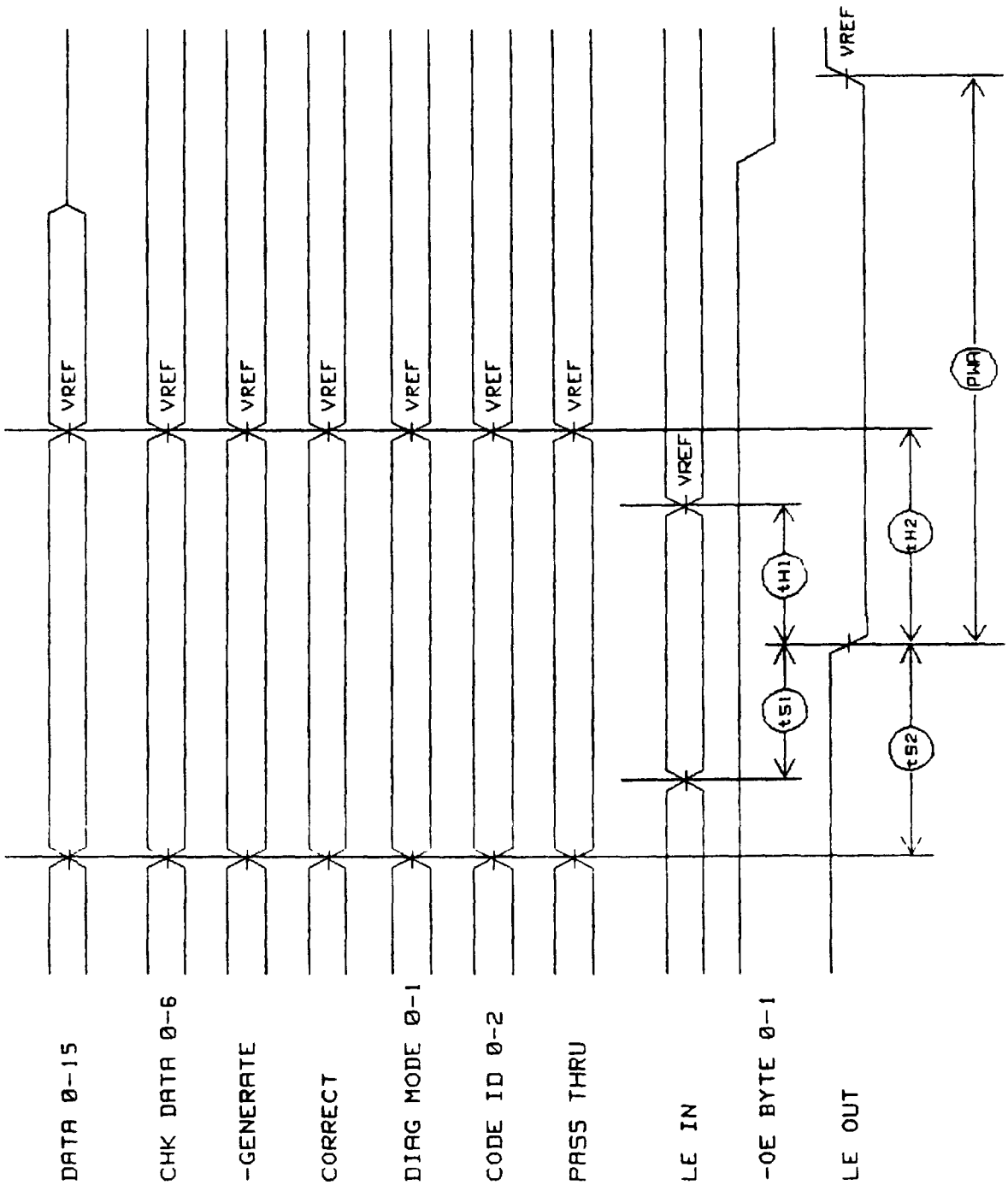


FIGURE 8. Input timing parameters for table XV.

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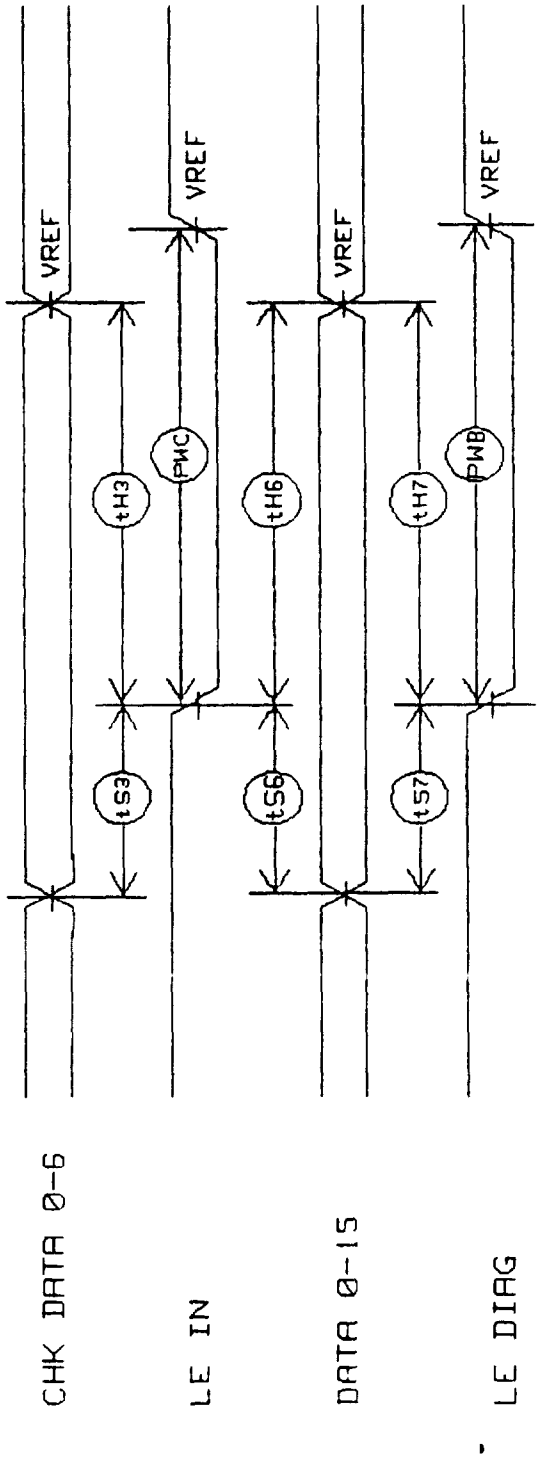


FIGURE 8. Input timing parameters for table XV - Continued.

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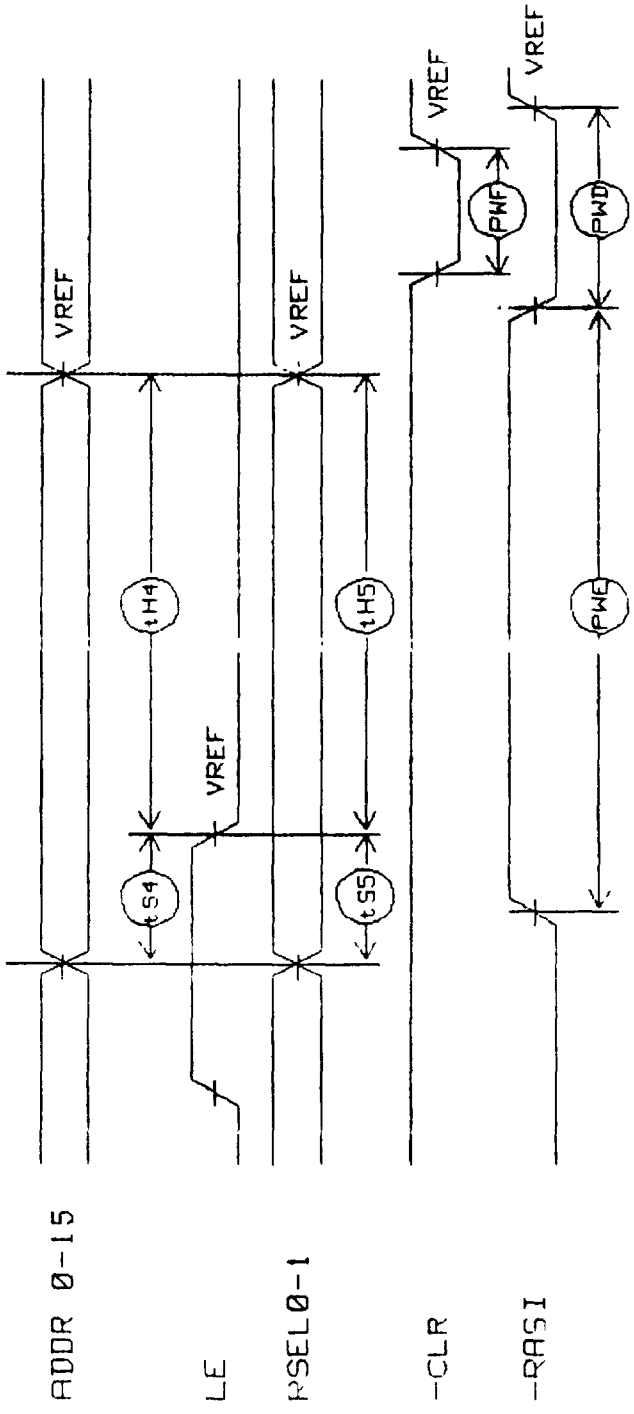


FIGURE 8. Input timing parameters for table XV - Continued

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NOTES:

1. Input parameters for figure 8.

PWA (LE OUT) = 15 nS minimum.
PWB (LE DIAG) = 15 nS minimum.
PWC (LE IN) = 15 nS minimum.
PWD (-RASI) = 50 nS minimum.
PWE (-RASI) = 50 nS minimum.
PWF (-CLR) = 35 nS minimum.
tS1 (LE IN) = 59 nS minimum.
tH1 (LE IN) = 5 nS minimum.
tS2 (DATA 0-15) = 50 nS minimum.
tS2 (CHK DATA 0-6, CODE ID 2-0 000,011) = 38 nS minimum.
tS2 (CHK DATA 0-6, CODE ID 2-0 010,100,101,110,111) = 30 nS minimum.
tS2 (-GENERATE) = 46 nS minimum.
tS2 (CORRECT) = 28 nS minimum.
tS2 (DIAG MODE 0-1) = 84 nS minimum.
tS2 (CODE ID 0-2) = 89 nS minimum.
tS2 (PASS THRU) = 30 nS minimum.

tH2 (DATA 0-15) = 5 nS minimum.
tH2 (CHK DATA 0-6, CODE ID 2-0 000,011) = 0 nS minimum.
tH2 (CHK DATA 0-6, CODE ID 2-0 010,100,101,110,111) = 0 nS minimum.
tH2 (-GENERATE) = 0 nS minimum.
tH2 (CORRECT) = 1 nS minimum.
tH2 (DIAG MODE 0-1) = 0 nS minimum.
tH2 (CODE ID 0-2) = 0 nS minimum.
tH2 (PASS THRU) = 0 nS minimum.

tS3 (CHK DATA 0-6) = 5 nS minimum.

tH3 (CHK DATA 0-6) = 7 nS minimum.

tS4 (ADDR 0-15) = 5 nS minimum.

tH4 (ADDR 0-15) = 15 nS minimum.

tS5 (RSEL0-1) = 5 nS minimum.

tH5 (RSEL0-1) = 25 nS minimum.

tS6 (DATA 0-15) = 7 nS minimum.

tH6 (DATA 0-15) = 7 nS minimum.

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NOTES: - Continued.

tS7 (DATA 0-15) = 7 nS minimum.

tH7 (DATA 0-15) = 9 nS minimum.

VREF = 1.5 ± 0.1 V.

2. Transition times (as measured between 0.8 V and 2.0 V points) shall not exceed 20 nS.

3. Pattern voltages shall be as specified in the test tolerance table.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-12
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-12
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-12
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-12

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	1.24 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105 mV	Q1,Q3
high, (Voh)	NA	± 155 mV	Q2,Q4
low, (Vol)	NA	± 25 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: 2.4 V ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ 0.5 V

3. The load circuit for this test is specified on figure 7.

4. All of the input waveforms shall meet the requirements shown on figure 9.

5. Input/Output (I/O) waveforms are as referenced on figure 10.

6. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

7. Floating inputs may cause damage to the module. All inputs should be tied high (3.0 V) or low (0.0 V) during testing.

8. DADDR 0-7 to -RAS 0-3 (-RFSH = H) skew is the maximum difference between fastest -RASI to -RAS 0-3 delay and slowest ADDR 0-15 to DADDR 0-7 within a single device.

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NOTES: - Continued.

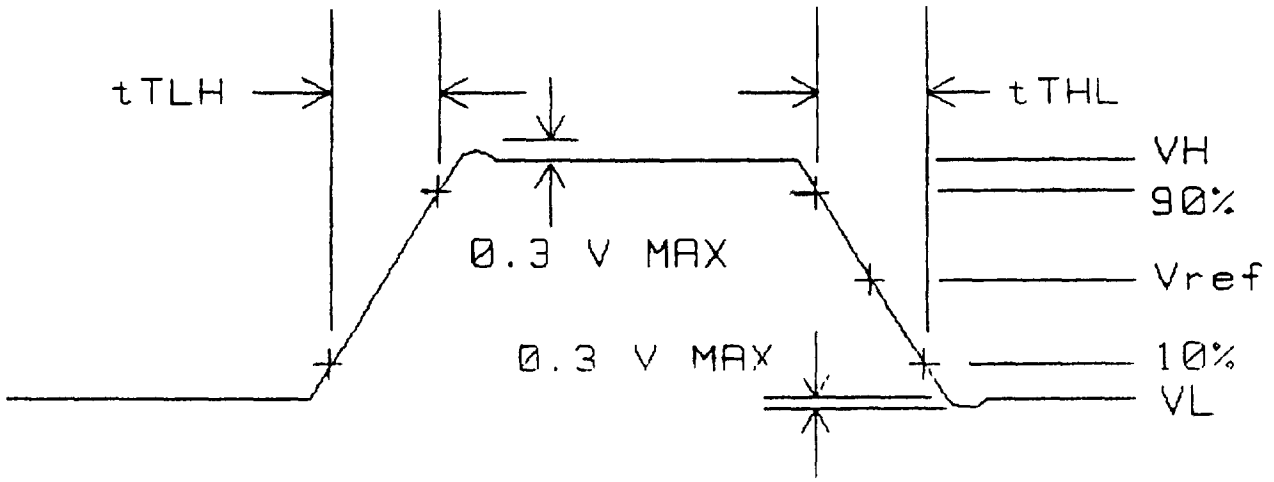
9. DADDR 0-7 to -CASO skew is the maximum difference between fastest -CASI to -CASO delay and slowest MSEL to DADDR 0-7 delay within a single device.

10. DADDR 0-7 to -RAS 0-3 (-RFSH = L) skew is the maximum difference between fastest -RASI to -RAS 0-3 delay and slowest -RFSH to DADDR 0-7 delay within a single device.

11. DADDR 0-7 to -RAS 0-3 (MSEL = falling edge) skew is the maximum difference between fastest MSEL falling edge to DADDR 0-7 delay and slowest -RASI to -RAS 0-3 delay within a single device.

12. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. Vref = 1.5 V.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
43,96	2.5 ± 1.5 nS	2.5 ± 1.5 nS
52,53,54,56,57,58,59,61,11, 12,63,64,66,67,17,68,69,70, 79,29,30,81,51,2,62,13,14,65, 16,28,21,22,27,23,24,76,20, 100,50,99,49,98,47,44,42,41, 89,38,33,80,83,35,88,39,90, 91,92,93,94,45,97	15.0 ± 1.5 nS	6.0 ± 1.5 nS

FIGURE 9. Sample input waveform for table XVI.

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TABLE XVII. Dynamic parameters.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
3	1-7	DATA 0-15 to SC 0-6 (tPHL)	-	-	-	46	46	46	nS	10a
3	8-14	DATA 0-15 to SC 0-6 (tPLH)	-	-	-	46	46	46	nS	10a
8	15-21	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 000,011 (tPHL)	-	-	-	41	41	41	nS	10a
8	22-28	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 000,011 (tPLH)	-	-	-	41	41	41	nS	10a
8	29-35	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 010,100, 110,111 (tPHL)	-	-	-	41	41	41	nS	10a
8	36-42	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 010,100, 110,111 (tPLH)	-	-	-	41	41	41	nS	10a
8	43	-GENERATE to SC 0-6 (tPHL)	-	-	-	49	49	49	nS	10a
8	44	-GENERATE to SC 0-6 (tPLH)	-	-	-	49	49	49	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		Not internal control mode:								
8	45-48	DIAG MODE 0-1 to SC 0-6 (tPHL)	-	-	-	69	69	69	nS	10a
8	49-52	DIAG MODE 0-1 to SC 0-6 (tPLH)	-	-	-	69	69	69	nS	10a
8	53	PASS THRU to SC 0-6 (tPHL)	-	-	-	50	50	50	nS	10a
8	54	PASS THRU to SC 0-6 (tPLH)	-	-	-	50	50	50	nS	10a
8	55-62	CODE ID to SC 0-6 (tPHL)	-	-	-	80	80	80	nS	10a
8	63-70	CODE ID to SC 0-6 (tPLH)	-	-	-	80	80	80	nS	10a
		From latched to trans-parent:								
4	71	LE IN to SC 0-6 (tPHL)	-	-	-	55	55	55	nS	10a
4	72	LE IN to SC 0-6 (tPLH)	-	-	-	55	55	55	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
4	73	LE DIAG to SC 0-6 (Not internal control mode) (tPHL)	-	-	-	61	61	61	nS	10a
4	74	LE DIAG to SC 0-6 (Not internal control mode) (tPLH)	-	-	-	61	61	61	nS	10a
		Internal control mode:								
4	75	LE DIAG to SC 0-6 (From latched to transparent) (tPHL)	-	-	-	86	86	86	nS	10a
4	76	LE DIAG to SC 0-6 (From latched to transparent) (tPLH)	-	-	-	86	86	86	nS	10a
8	77-83	DATA 0-15 to SC 0-6 (Via diagnostic latch) (tPHL)	-	-	-	86	86	86	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	84-90	DATA 0-15 to SC 0-6 (Via diagnostic latch) (tPLH)	-	-	-	86	86	86	nS	10a
14	91-106	DATA 0-15 In to correct DATA 0-15 Out (tPHL)	-	-	-	84	84	84	nS	10c
14	107-122	DATA 0-15 In to correct DATA 0-15 Out (tPLH)	-	-	-	84	84	84	nS	10c
8	123-138	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 000, 011) (tPHL)	-	-	-	72	72	72	nS	10a
8	139-154	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 000, 011) (tPLH)	-	-	-	72	72	72	nS	10a
8	155-170	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 010, 100,101, 110,111) (tPHL)	-	-	-	61	61	61	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	171-186	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 010, 100,101, 110,111) (tPLH)	-	-	-	61	61	61	nS	10a
8	187	-GENERATE to DATA 0-15 (tPLH)	-	-	-	80	80	80	nS	10a
8	188	-GENERATE to DATA 0-15 (tPLH)	-	-	-	80	80	80	nS	10a
		Not internal control mode:								
8	189	CORRECT to DATA 0-15 (tPHL)	-	-	-	60	60	60	nS	10a
8	190	CORRECT to DATA 0-15 (tPLH)	-	-	-	60	60	60	nS	10a
8	191-194	DIAG MODE 0-1 to DATA 0-15 (tPHL)	-	-	-	100	100	100	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		Not internal control mode:								
8	195-198	DIAG MODE to DATA 0-15 (tPLH)	-	-	-	100	100	100	nS	10a
8	199	PASS THRU to DATA 0-15 (tPHL)	-	-	-	62	62	62	nS	10a
8	200	PASS THRU to DATA 0-15 (tPLH)	-	-	-	62	62	62	nS	10a
8	201-208	CODE ID to DATA 0-15 (tPHL)	-	-	-	111	111	111	nS	10a
8	209-216	CODE ID to DATA 0-15 (tPLH)	-	-	-	111	111	111	nS	10a
		From latched to trans-parent:								
18	217	LE IN to Correct DATA 0-15 Out (tPHL)	-	-	-	93	93	93	nS	10d
18	218	LE IN to Correct DATA 0-15 Out (tPLH)	-	-	-	93	93	93	nS	10d

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
16	219	LE OUT to DATA 0-15 (tPHL)	-	-	-	44	44	44	nS	10c
16	220	LE OUT to DATA 0-15 (tPLH)	-	-	-	44	44	44	nS	10c
19	221	LE DIAG to DATA 0-15 (Not internal control mode) (tPHL)	-	-	-	99	99	99	nS	10d
19	222	LE DIAG to DATA 0-15 (Not internal control mode) (tPLH) Internal control mode:	-	-	-	99	99	99	nS	10d
19	223	LE DIAG to DATA 0-15 (From latched to transparent) (tPHL)	-	-	-	117	117	117	nS	10d
19	224	LE DIAG to DATA 0-15 (From latched to transparent) (tPLH)	-	-	-	117	117	117	nS	10d

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
14	225-240	DATA to DATA 0-15 (Via diagnostic latch) (tPHL)	-	-	-	117	117	117	nS	10c
14	241-256	DATA to DATA 0-15 (Via diagnostic latch) (tPLH)	-	-	-	117	117	117	nS	10c
3	257-272	DATA 0-15 to -ERROR (tPHL)	-	-	-	47	47	47	nS	10a
3	273-288	DATA 0-15 to -ERROR (tPLH)	-	-	-	47	47	47	nS	10a
8	289-295	CHK DATA 0-6 to -ERROR (CODE ID 2-0 000, 011) (tPHL)	-	-	-	42	42	42	nS	10a
8	296-302	CHK DATA 0-6 to -ERROR (CODE ID 2-0 000, 011) (tPLH)	-	-	-	42	42	42	nS	10a
8	303-309	CHK DATA 0-6 to -ERROR (CODE ID 2-0 100, 110,111) (tPHL)	-	-	-	42	42	42	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	310-316	CHK DATA 0-6 to -ERROR (CODE ID 2-0 100, 110,111) (tPLH)	-	-	-	42	42	42	nS	10a
8	317	-GENERATE to -ERROR (tPHL)	-	-	-	52	52	52	nS	10a
8	318	-GENERATE to -ERROR (tPLH)	-	-	-	52	52	52	nS	10a
		Not internal control mode:								
8	319	DIAG MODE 0-1 to -ERROR (tPHL)	-	-	-	76	76	76	nS	10a
8	320	DIAG MODE 0-1 to -ERROR (tPLH)	-	-	-	76	76	76	nS	10a
8	321	PASS THRU to -ERROR (tPHL)	-	-	-	45	45	45	nS	10a
8	322	PASS THRU to -ERROR (tPLH)	-	-	-	45	45	45	nS	10a
8	323-325	CODE ID to -ERROR (tPHL)	-	-	-	79	79	79	nS	10a
8	326-328	CODE ID to -ERROR (tPLH)	-	-	-	79	79	79	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		From latched to transparent:								
4	329	LE IN to -ERROR (tPHL)	-	-	-	54	54	54	nS	10a
4	330	LE IN to -ERROR (tPLH)	-	-	-	54	54	54	nS	10a
4	331	LE DIAG to -ERROR (Not internal control mode) (tPHL)	-	-	-	60	60	60	nS	10a
4	332	LE DIAG to -ERROR (Not internal control mode) (tPLH)	-	-	-	60	60	60	nS	10a
		Internal control mode:								
4	333	LE DIAG to -ERROR (From latched to transparent) (tPHL)	-	-	-	85	85	85	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
4	334	LE DIAG to -ERROR (From latched to trans-parent) (tPLH)	-	-	-	85	85	85	nS	10a
8	335-350	DATA 0-15 to -ERROR (Via di-agnostic latch) (tPHL)	-	-	-	85	85	85	nS	10a
8	351-366	DATA 0-15 to -ERROR (Via di-agnostic latch) (tPLH)	-	-	-	85	85	85	nS	10a
3	367-382	DATA 0-15 to -MULTI ERROR (tPHL)	-	-	-	67	67	67	nS	10a
3	383-398	DATA 0-15 TO -MULTI ERROR (tPLH)	-	-	-	67	67	67	nS	10a
8	399-414	CHK DATA 0-6 to -MULTI ERROR (CODE ID 2-0 000, 011) (tPHL)	-	-	-	61	61	61	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	415-430	CHK DATA 0-6 to -MULTI ERROR (CODE ID 2-0 000, 011) (tPLH)	-	-	-	61	61	61	nS	10a
8	431-437	CHK DATA 0-6 to -MULTI ERROR (CODE ID 2-0 100, 110,111) (tPHL)	-	-	-	48	48	48	nS	10a
8	438-444	CHK DATA 0-6 to -MULTI ERROR (CODE ID 2-0 100, 110,111) (tPLH)	-	-	-	48	48	48	nS	10a
8	445	-GENERATE to -MULTI ERROR (tPHL)	-	-	-	73	73	73	nS	10a
8	446	-GENERATE to -MULTI ERROR (tPLH)	-	-	-	73	73	73	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		Not internal control mode:								
8	447-450	DIAG MODE 0-1 to -MULTI ERROR (tPHL)	-	-	-	101	101	101	nS	10a
8	451-454	DIAG MODE 0-1 to -MULTI ERROR (tPLH)	-	-	-	101	101	101	nS	10a
8	455	PASS THRU to -MULTI ERROR (tPHL)	-	-	-	65	65	65	nS	10a
8	456	PASS THRU to -MULTI ERROR (tPLH)	-	-	-	65	65	65	nS	10a
8	457-460	CODE ID to -MULTI ERROR (tPHL)	-	-	-	101	101	101	nS	10a
8	461-464	CODE ID to -MULTI ERROR (tPLH)	-	-	-	101	101	101	nS	10a
		From latched to trans-parent:								
4	465	LE IN to -MULTI ERROR (tPHL)	-	-	-	77	77	77	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
4	466	LE IN to -MULTI ERROR (tPLH)	-	-	-	77	77	77	nS	10a
4	467	LE DIAG to -MULTI ERROR (Not internal control mode) (tPHL)	-	-	-	83	83	83	nS	10a
4	468	LE DIAG to -MULTI ERROR (Not internal control mode) (tPLH)	-	-	-	83	83	83	nS	10a
		Internal control mode:								
4	469	LE DIAG to -MULTI ERROR (From latched to transparent) (tPHL)	-	-	-	107	107	107	nS	10a
4	470	LE DIAG to -MULTI ERROR (From latched to transparent) (tPLH)	-	-	-	107	107	107	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	471-486	DATA 0-15 to -MULTI ERROR (Via diagnostic latch) (tPHL)	-	-	-	107	107	107	nS	10a
8	487-502	DATA 0-15 to -MULTI ERROR (Via diagnostic latch) (tPLH)	-	-	-	107	107	107	nS	10a
1	503	DATA 0-15 to LE IN setup time (tS)	7	7	7	-	-	-	nS	10a 10c
2	504	DATA 0-15 to LE IN hold time (tH)	7	7	7	-	-	-	nS	10a 10c
1	505	CHK DATA 0-6 to LE IN setup time (tS)	5	5	5	-	-	-	nS	10a 10c
2	506	CHK DATA 0-6 to LE IN hold time (tH)	7	7	7	-	-	-	nS	10a
10	507	DATA 0-15 to LE OUT setup time (tS)	50	50	50	-	-	-	nS	10b
11	508	DATA 0-15 to LE OUT hold time (tH)	5	5	5	-	-	-	nS	10b

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TABLE XVII. Dynamic parameters - Continued.

ef im	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
10	509	CHK DATA 0-6 to LE OUT (CODE ID 2-0 000,011) setup time (tS)	38	38	38	-	-	-	nS	10b
.1	510	CHK DATA 0-6 to LE OUT (CODE ID 2-0 010,011) hold time (tH)	0	0	0	-	-	-	nS	10b
0	511	CHK DATA 0-6 to LE OUT (CODE ID 2-0 010,100, 101,110, 111) set-up time (tS)	30	30	30	-	-	-	nS	10b
1	512	CHK DATA 0-6 to LE OUT (CODE ID 2-0 010,100, 101,110, 111) hold time (tH)	0	0	0	-	-	-	nS	10b
0	513	-GENERATE to LE OUT setup time (tS)	46	46	46	-	-	-	nS	10b
	514	-GENERATE to LE OUT hold time (tH)	0	0	0	-	-	-	nS	10b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
10	515	CORRECT to LE OUT setup time (tS)	28	28	28	-	-	-	nS	10b
11	516	CORRECT to LE OUT hold time (tH)	1	1	1	-	-	-	nS	10b
10	517	DIAG MODE 0-1 to LE OUT setup time (tS)	84	84	84	-	-	-	nS	10b
11	518	DIAG MODE 0-1 to LE OUT hold time (tH)	0	0	0	-	-	-	nS	10b
10	519	PASS THRU to LE OUT setup time (tS)	30	30	30	-	-	-	nS	10b
11	520	PASS THRU to LE OUT hold time (tH)	0	0	0	-	-	-	nS	10b
10	521	CODE ID to LE OUT setup time (tS)	89	89	89	-	-	-	nS	10b
11	522	CODE ID to LE OUT hold time (tH)	0	0	0	-	-	-	nS	10b
12	523	LE IN to LE OUT setup time (tS)	59	59	59	-	-	-	nS	10b
13	524	LE IN to LE OUT hold time (tH)	5	5	5	-	-	-	nS	10b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
1	525	DATA 0-15 to LE DIAG set-up time (tS)	7	7	7	-	-	-	nS	10a 10c
2	526	DATA 0-15 to LE DIAG hold time (tH)	9	9	9	-	-	-	nS	10a
5	527-528	-OE BYTE 0-1 to DATA 0-15 (tPZH)	-	-	-	46	46	46	nS	10a
5	529-530	-OE BYTE 0-1 to DATA 0-15 (tPZL)	-	-	-	46	46	46	nS	10a
7	531	-OE SC to SC 0-6 (tPZH)	-	-	-	46	46	46	nS	10a
7	532	-OE SC to SC 0-6 (tPZL)	-	-	-	46	46	46	nS	10a
1	536-551	ADDR 0-15 to DADDR 0-7 (tPHL)	-	-	-	51	46	51	nS	10e
1	552-567	ADDR 0-15 to DADDR 0-7 (tPLH)	-	-	-	50	45	50	nS	10e
2	568	-RASI to -RAS 0-3 (-RFSH=H) (tPHL)	-	-	-	51	46	51	nS	10e
2	569	-RASI to -RAS 0-3 (-RFSH=H) (tPLH)	-	-	-	52	47	52	nS	10e

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
3	570	-RASI to -RAS 0-3 (-RFSH=L) (tPHL)	-	-	-	51	46	51	nS	10e
3	571	-RASI to -RAS 0-3 (-RFSH=L) (tPLH)	-	-	-	52	47	52	nS	10e
4	572	MSEL to DADDR 0-7 Row address invalid (tPHL)	9	9	9	53	48	53	nS	10e
4	573	MSEL to DADDR 0-7 Row address invalid (tPLH)	9	9	9	52	47	52	nS	10e
5	574-575	MSEL to DADDR 0-7 Column address valid (tPHL)	9	9	9	53	48	53	nS	10e
5	576-577	MSEL to DADDR 0-7 Column address valid (tPLH)	9	9	9	52	47	52	nS	10e
6	578	-CASI to -CASO (-RFSH=H) (tPHL)	-	-	-	25	25	25	nS	10e
6	579	-CASI to -CASO (-RFSH=H) (tPLH)	-	-	-	31	31	31	nS	10e

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
7	580-583	RSEL0-1 to -RAS 0-3 (LE= H, -RASI= L) (tPHL)	-	-	-	52	47	52	nS	10e
7	584-587	RSEL0-1 to -RAS 0-3 (LE= H, -RASI= L) (tPLH)	-	-	-	51	46	51	nS	10e
8	588	-RFSH to -TC (-RASI=L) (tPLH)	-	-	-	56	56	56	nS	10e
9	589	-RASI to -TC (-RFSH=L) (tPLH)	-	-	-	46	46	46	nS	10e
10	590	-RASI pulse width low (-RFSH=L) (PWL)	50	50	50	-	-	-	nS	10e
11	591	-RASI pulse width high (-RFSH=L) (PWH)	50	50	50	-	-	-	nS	10e
12	592	-RFSH to DADDR 0-7 refresh address valid (-RASI=X) (tPHL)	-	-	-	53	48	53	nS	10e

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
12	593	-RFSH to DADDR 0-7 refresh address valid (-RASI=X) (tPLH)	-	-	-	52	47	52	nS	10e
13	594	-RFSH to -RAS 0-3 (-RASI=L) (tPHL)	-	-	-	57	57	57	nS	10e
14	595	-CLR (PWL)	35	35	35	-	-	-	nS	10e
15	596	-RFSH to -CASO (tPLH) (-RASI=L, -CASI=L)	-	-	-	31	31	31	nS	10e
16	597	LE to DADDR 0-7 valid (tPHL)	-	-	-	68	63	68	nS	10e
16	598	LE to DADDR 0-7 valid (tPLH)	-	-	-	67	62	67	nS	10e
17	599	LE to -RAS 0-3 (tPHL)	-	-	-	73	68	73	nS	10e
18	600	-CLR to -TC	-	-	-	62	62	62	nS	10e
19	601	-CLR to DADDR 0-7 valid (-RFSH=L) (tPLH)	-	-	-	81	76	81	nS	10e
20	602-617	ADDR 0-15 to LE (HL) set-up time (tS)	5	5	5	-	-	-	nS	10e

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
21	618-633	ADDR 0-15 to LE (HL) hold time (tH)	15	15	15	-	-	-	nS	10e
22	634-635	RSEL0-1 to LE (HL) set-up time (tS)	5	5	5	-	-	-	nS	10e
23	636-637	RSEL0-1 to LE (HL) hold time (tH)	25	25	25	-	-	-	nS	10e
24	638	-CLR recovery time	18	18	18	-	-	-	nS	10e
25	639-646	DADDR 0-7 to -RAS 0-3 skew time (-RFSH=H) (tSKEW)	-	-	-	12	12	12	nS	10e
26	647-654	DADDR 0-7 to -CASO (tSKEW)	-	-	-	14	14	14	nS	10e
27	655-662	DADDR 0-7 to -RAS 0-3 (-RFSH=L) (tSKEW)	-	-	-	16	16	16	nS	10e
28	663-670	DADDR 0-7 to -RAS 0-3 (MSEL =Falling EDGE) (tSKEW)	-	-	-	11	11	11	nS	10e
1	671	-CAS IN to -CAS OUT A-B (tPHL)	-	-	-	28	23	28	nS	10f

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
2	672	-CAS IN to -CAS OUT A-B (tPLH)	-	-	-	27	22	27	nS	10f
3	673	-WE IN to -WE OUT A-B (tPHL)	-	-	-	28	23	28	nS	10f
4	674	-WE IN to -WE OUT A-B (tPLH)	-	-	-	27	22	27	nS	10f
3	675-681	DATA 0-15 to SC 0-6 (tPHL)	-	-	-	62	62	62	nS	10a
3	682-688	DATA 0-15 to SC 0-6 (tPLH)	-	-	-	62	62	62	nS	10a
8	689-695	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 000,011 (tPHL)	-	-	-	57	57	57	nS	10a
8	696-702	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 000,011 (tPLH)	-	-	-	57	57	57	nS	10a
8	703-709	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 010,100, 101,110, 111 (tPHL)	-	-	-	57	57	57	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	710-716	CHK DATA 0-6 to SC 0-6 CODE ID 2-0 010,100, 101,110, 111 (tPLH)	-	-	-	57	57	57	nS	10a
8	717	-GENERATE to SC 0-6 (tPHL)	-	-	-	65	65	65	nS	10a
8	718	-GENERATE to SC 0-6 (tPLH)	-	-	-	65	65	65	nS	10a
		Not internal control mode:								
8	719-722	DIAG MODE 0-1 to SC 0-6 (tPHL)	-	-	-	85	85	85	nS	10a
8	723-726	DIAG MODE 0-1 to SC 0-6 (tPLH)	-	-	-	85	85	85	nS	10a
8	727	PASS THRU to SC 0-6 (tPHL)	-	-	-	66	66	66	nS	10a
8	728	PASS THRU to SC 0-6 (tPLH)	-	-	-	66	66	66	nS	10a
8	729-736	CODE ID 0-2 to SC 0-6 (tPHL)	-	-	-	96	96	96	nS	10a
8	737-744	CODE ID 0-2 to SC 0-6 (tPLH)	-	-	-	96	96	96	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
4	745	LE IN to SC 0-6 (From latched to trans-parent) (tPHL) From latched to trans-parent:	-	-	-	71	71	71	nS	10a
4	746	LE IN to SC 0-6 (tPLH)	-	-	-	71	71	71	nS	10a
4	747	LE DIAG to SC 0-6 (Not internal control mode) (tPHL)	-	-	-	77	77	77	nS	10a
4	748	LE DIAG to SC 0-6 (Not internal control mode) (tPLH)	-	-	-	77	77	77	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		Internal control mode:								
4	749	LE DIAG to SC 0-6 (From latched to trans-parent) (tPHL)	-	-	-	102	102	102	nS	10a
4	750	LE DIAG to SC 0-6 (From latched to trans-parent) (tPLH)	-	-	-	102	102	102	nS	10a
8	751-758	DATA 0-15 to SC 0-6 (Via di-agnostic latch) (tPHL)	-	-	-	102	102	102	nS	10a
8	759-766	DATA 0-15 to SC 0-6 (Via di-agnostic latch) (tPLH)	-	-	-	102	102	102	nS	10a
14	767-782	DATA 0-15 In to cor-rect DATA 0-15 Out (tPHL)	-	-	-	100	100	100	nS	10c
14	783-798	DATA 0-15 In to cor-rect DATA 0-15 Out (tPLH)	-	-	-	100	100	100	nS	10c

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
8	799-805	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 000, 011) (tPHL)	-	-	-	88	88	88	nS	10a
8	806-812	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 000, 011) (tPLH)	-	-	-	88	88	88	nS	10a
8	813-819	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 010, 100,101 110,111) (tPHL)		-	-	77	77	77	nS	10a
8	820-826	CHK DATA 0-6 to DATA 0-15 (CODE ID 2-0 010, 100,101, 110,111) (tPHL)	-	-	-	77	77	77	nS	10a
8	827	-GENERATE to DATA 0-15 (tPHL)	-	-	-	96	96	96	nS	10a
8	828	-GENERATE to DATA 0-15 (tPLH)	-	-	-	96	96	96	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		Not internal control mode:								
8	829	CORRECT to DATA 0-15 (tPHL)	-	-	-	76	76	76	nS	10a
8	830	CORRECT to DATA 0-15 (tPLH)	-	-	-	76	76	76	nS	10a
8	831-834	DIAG MODE 0-1 to DATA 0-15 (tPHL)	-	-	-	116	116	116	nS	10a
8	835-838	DIAG MODE 0-1 to DATA 0-15 (tPLH)	-	-	-	116	116	116	nS	10a
8	839	PASS THRU to DATA 0-15 (tPHL)	-	-	-	78	78	78	nS	10a
8	840	PASS THRU to DATA 0-15 (tPLH)	-	-	-	78	78	78	nS	10a
8	841-848	CODE ID 0-2 to DATA 0-15 (tPHL)	-	-	-	127	127	127	nS	10a
8	849-856	CODE ID 0-2 to DATA 0-15 (tPLH)	-	-	-	127	127	127	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		From latched to transparent:								
18	857	LE IN to correct DATA 0-15 Out (tPHL)	-	-	-	109	109	109	nS	10d
18	858	LE IN to correct DATA 0-15 Out (tPLH)	-	-	-	109	109	109	nS	10d
16	859	LE OUT to DATA 0-15 (tPHL)	-	-	-	60	60	60	nS	10c
16	860	LE OUT to DATA 0-15 (tPLH)	-	-	-	60	60	60	nS	10c
19	861	LE DIAG to DATA 0-15 (Not internal control mode) (tPHL)	-	-	-	115	115	115	nS	10d
19	862	LE DIAG to DATA 0-15 (Not internal control mode) (tPLH)	-	-	-	115	115	115	nS	10d

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
		Internal control mode:								
19	863	LE DIAG to DATA 0-15 (From latched to trans-parent) (tPHL)	-	-	-	133	133	133	nS	10d
19	864	LE DIAG to DATA 0-15 (From latched to trans-parent) (tPLH)	-	-	-	133	133	133	nS	10d
14	865-880	DATA 0-15 to DATA 0-15 (Via diagnostic latch) (tPHL)	-	-	-	133	133	133	nS	10c
14	881-896	DATA 0-15 to DATA 0-15 (Via diagnostic latch) (tPHL)	-	-	-	133	133	133	nS	10c
5	897-898	-OE BYTE 0-1 to DATA 0-15 (tPZH)	-	-	-	62	62	62	nS	10a
5	899-900	-OE BYTE 0-1 to DATA 0-15 (tPZL)	-	-	-	62	62	62	nS	10a

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
7	901	-OE SC to SC 0-6 (tPZH)	-	-	-	62	62	62	nS	10a
7	902	-OE SC to SC 0-6 (tPZL)	-	-	-	62	62	62	nS	10a
1	903-918	ADDR 0-15 to DADDR 0-7 (tPHL)	-	-	-	61	56	61	nS	10e
1	919-934	ADDR 0-15 to DADDR 0-7 (tPLH)	-	-	-	63	58	63	nS	10e
2	935	-RASI to -RAS 0-3 (-RFSH=H) (tPHL)	-	-	-	61	56	61	nS	10e
2	936	-RASI to -RAS 0-3 (-RFSH=H) (tPLH)	-	-	-	65	58	65	nS	10e
3	937	-RASI to -RAS 0-3 (-RFSH=L) (tPHL)	-	-	-	61	56	61	nS	10e
3	938	-RASI to -RAS 0-3 (-RFSH=L) (tPLH)	-	-	-	65	58	65	nS	10e
4	939	MSEL to DADDR 0-7 (Row address invalid) (tPHL)	-	-	-	63	58	63	nS	10e
4	940	MSEL to DADDR 0-7 (Row address invalid) (tPLH)	-	-	-	65	60	65	nS	10e

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
5	941-942	MSEL to DADDR 0-7 Column address valid/invalid (tPHL)	-	-	-	63	58	63	nS	10e
5	943-944	MSEL to DADDR 0-7 Column address valid/invalid (tPLH)	-	-	-	65	60	65	nS	10e
7	945-948	RSEL0-1 to -RAS 0-3 (LE=H, -RASI=L) (tPHL)	-	-	-	62	57	62	nS	10e
7	949-952	RSEL0-1 to -RAS 0-3 (LE=H, -RASI=L) (tPLH)	-	-	-	64	59	64	nS	10e
12	953	-RFSH to DADDR 0-7 Row address valid (-RASI=X) (tPHL)	-	-	-	63	58	63	nS	10e
12	954	-RFSH to DADDR 0-7 Row address valid (-RASI=X) (tPLH)	-	-	-	65	60	65	nS	10e

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
13	955	-RFSH to -RAS 0-3 (-RASI=L) (tPHL)	-	-	-	67	62	67	nS	10e
16	956	LE to DADDR 0-7 Valid (tPHL)	-	-	-	78	73	78	nS	10e
16	957	LE to DADDR 0-7 Valid (tPLH)	-	-	-	80	75	80	nS	10e
17	958	LE to -RAS 0-3 (tPHL)	-	-	-	83	78	83	nS	10e
19	959	-CLR to DADDR 0-7 Valid (-RFSH=L) (tPLH)	-	-	-	94	89	94	nS	10e
1	960	-CAS IN to -CAS OUT A-B (tPHL)	-	-	-	38	33	38	nS	10f
2	961	-CAS IN to -CAS OUT A-B (tPLH)	-	-	-	40	35	40	nS	10f
3	962	-WE IN to -WE OUT A-B (tPHL)	-	-	-	38	33	38	nS	10f
4	963	-WE IN to -WE OUT A-B (tPLH)	-	-	-	40	35	40	nS	10f

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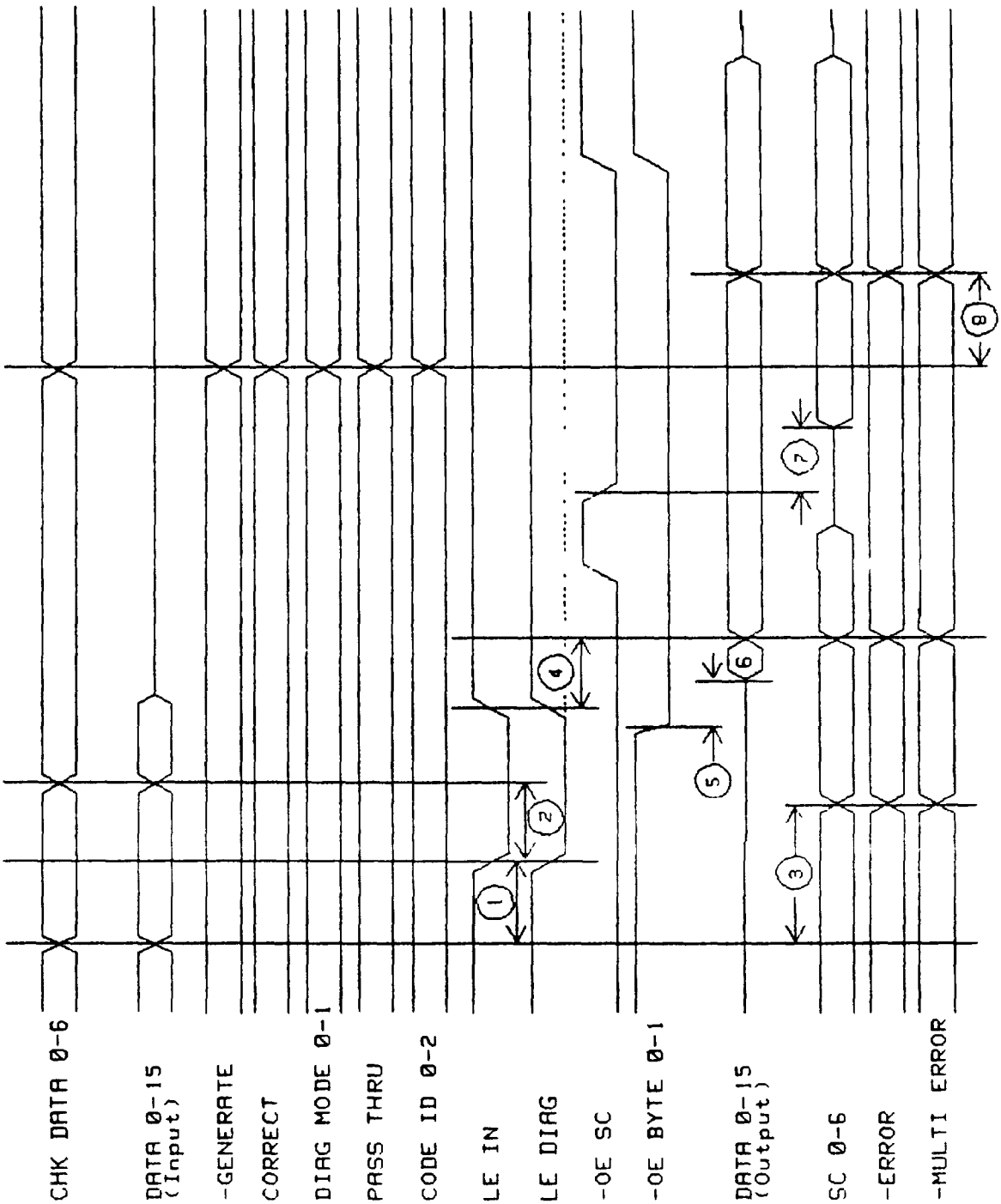


FIGURE 10a. Dynamic waveforms for table XVII.

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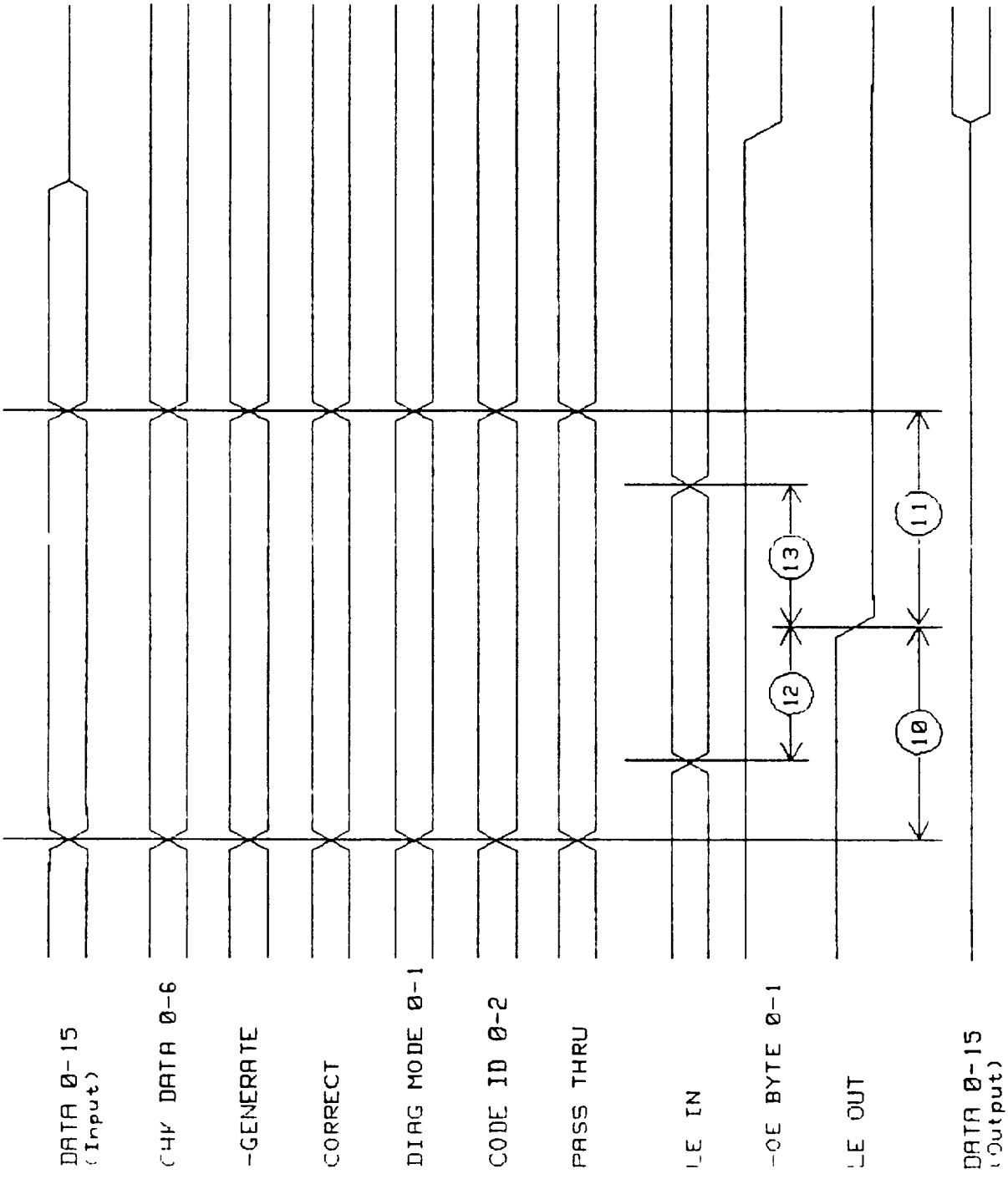


FIGURE 10b. Dynamic waveforms for table XVII.

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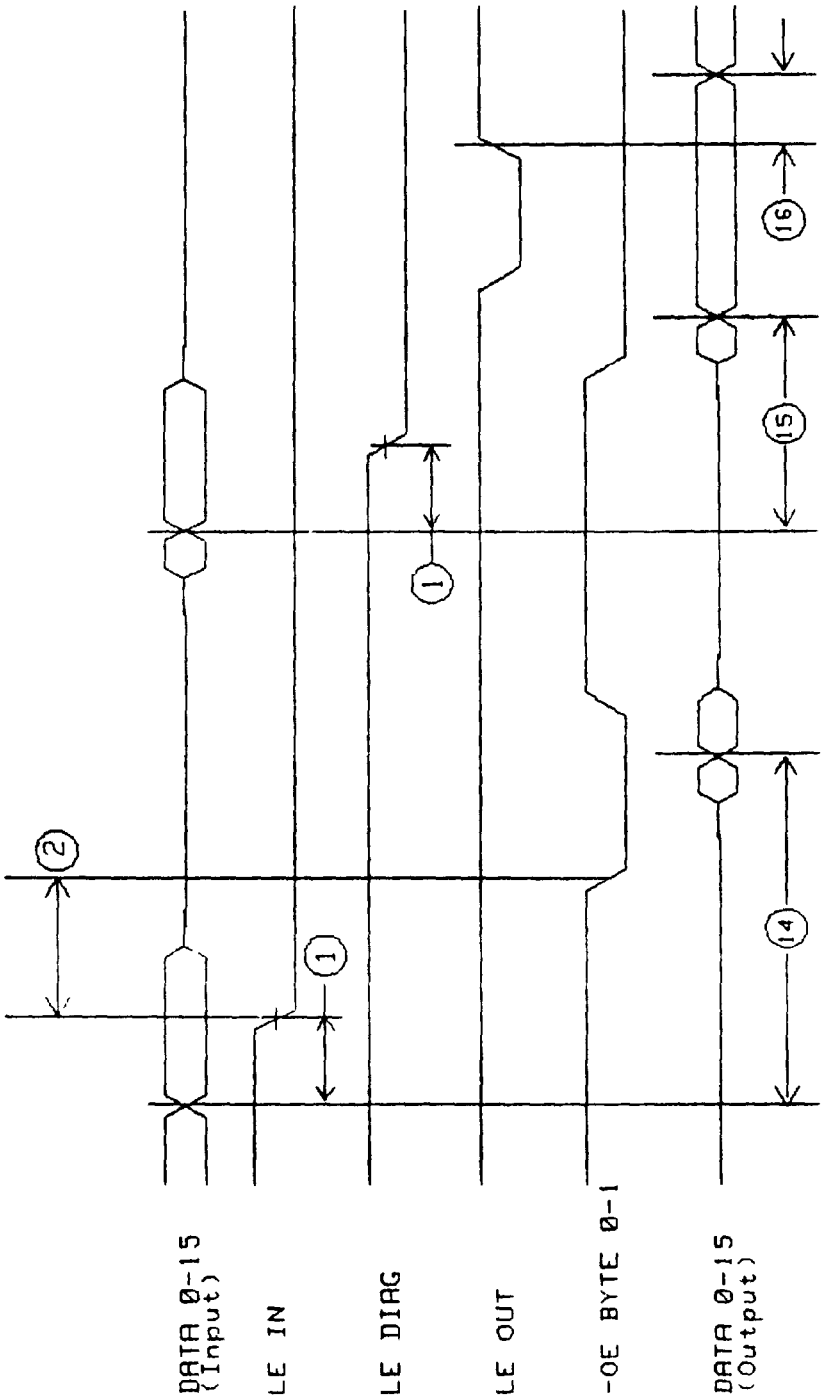


FIGURE 10c. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
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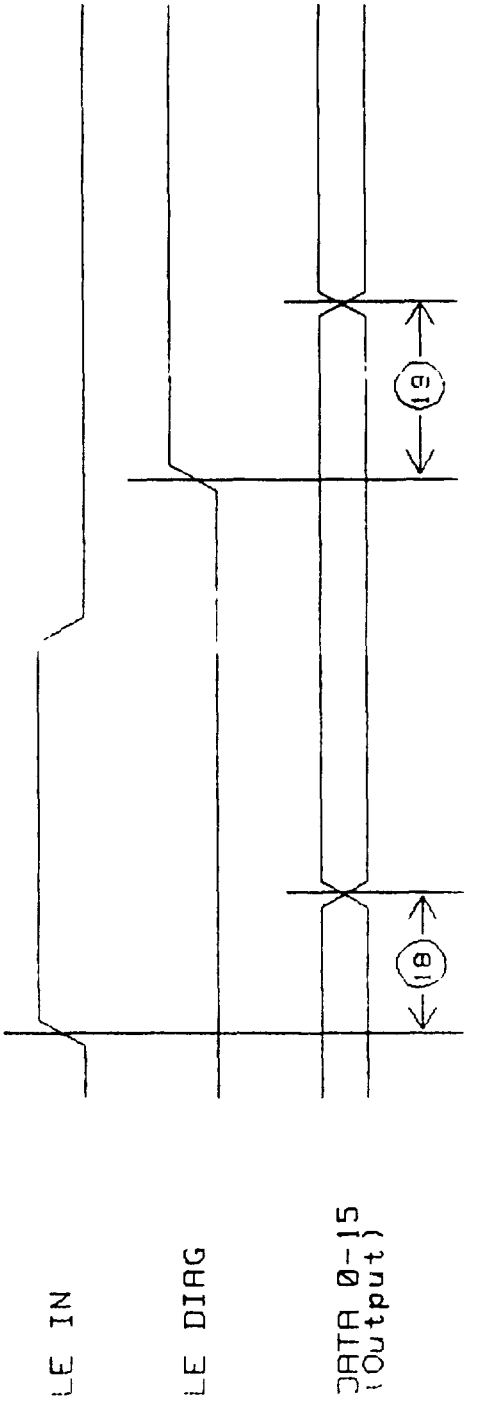


FIGURE 10d. Dynamic waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:04Z
Check the source to verify that this is the current version before use.

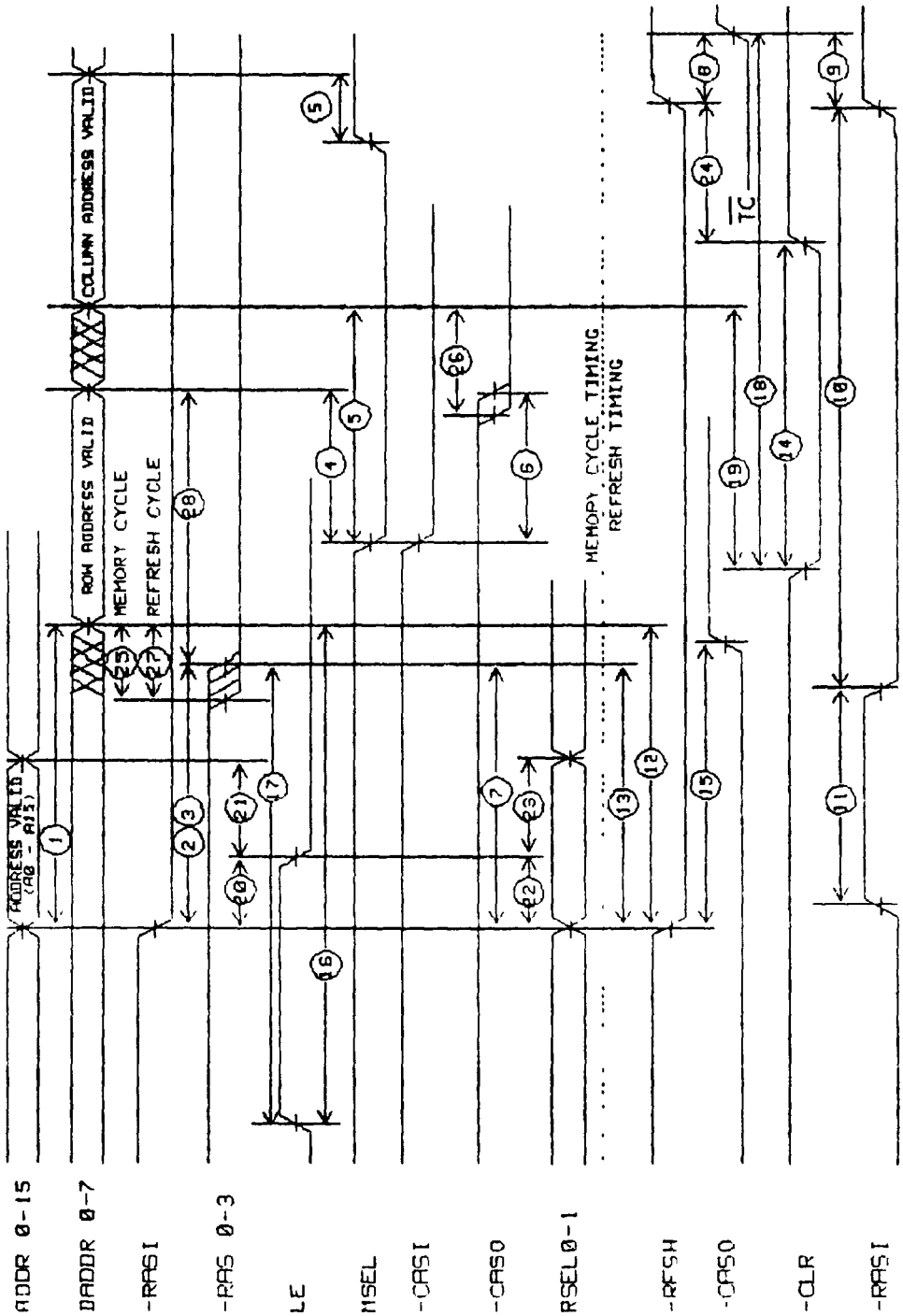


FIGURE 10e. Dynamic waveforms for table XVII.

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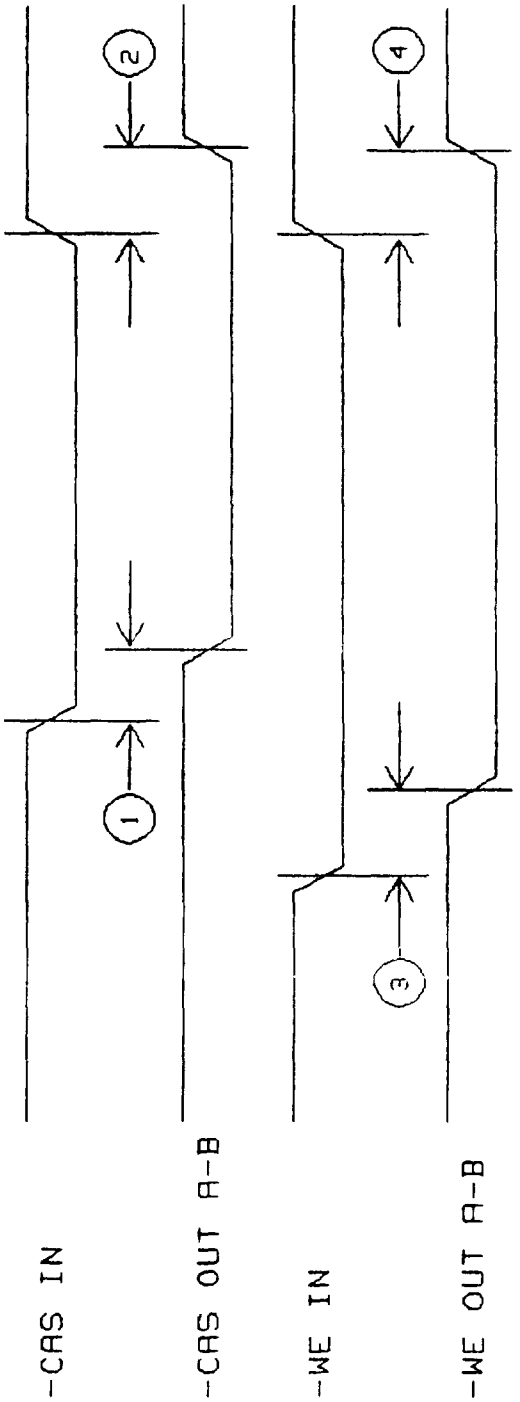


FIGURE 10f. Dynamic waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.6	6.6	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 130 nF	R

2 The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1	0	400	mV	1-5

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	1.24 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D07, inspect for the most negative spike as shown on figure 11 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

4. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

5. Abbreviations used are defined in 6.1.

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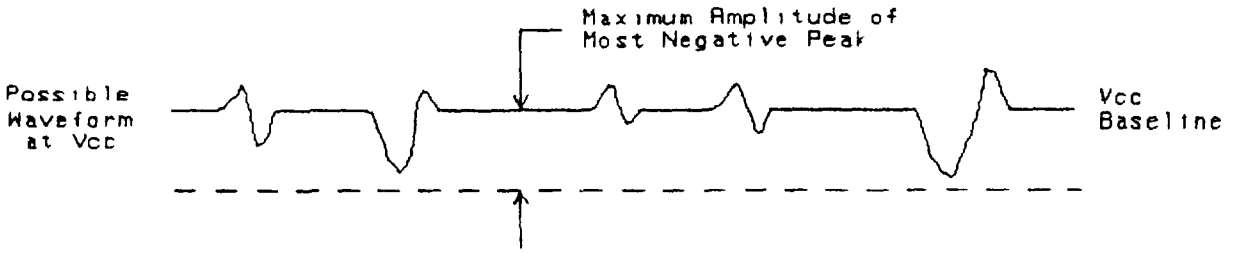


FIGURE 11. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
T1	SC 0	71	0	500	mV	1-7	12,13
T2	SC 1	26					
T3	SC 2	74					
T4	SC 3	73					
T5	SC 4	77					
T6	SC 5	72					
T7	SC 6	78					
T8	-TC	48					
T9	-CASO	46					
T10	DADDR 0	8					
T11	DADDR 1	9					
T12	DADDR 2	60					
T13	DADDR 3	4					
T14	DADDR 4	3					
T15	DADDR 5	5					
T16	DADDR 6	6					
T17	DADDR 7	7					
T18	-RAS 0	87					
T19	-RAS 1	86					
T20	-RAS 2	36	0	500	mV	1-7	12,13

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs.
T21	-RAS 3	37	0	500	mV	1-7	12,13
T22	-CAS OUT A	32					
T23	-CAS OUT B	82					
T24	-WE OUT A	84					
T25	-WE OUT B	34					
T26	-ERROR	19					
T27	-MULTI ERROR	18					
T28	DATA 0	81					
T29	DATA 1	30					
T30	DATA 2	29					
T31	DATA 3	79					
T32	DATA 4	68					
T33	DATA 5	17					
T34	DATA 6	67					
T35	DATA 7	66					
T36	DATA 8	64					
T37	DATA 9	63					
T38	DATA 10	12					
T39	DATA 11	11					
T40	DATA 12	57	0	500	mV	1-7	12,13

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs.
T41	DATA 13	56	0	500	mV	1-7	12,13
T42	DATA 14	54	|	|	|	|	|
T43	DATA 15	53	0	500	mV	1-7	12,13

NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T43
Power supply current limit	1.24 A	NA	T1-T43
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T43 T1-T43
Output crosstalk measurement	NA	± 25 mV	T1-T43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 12.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 13.

4. All outputs shall be loaded as specified on figure 7.

5. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

6. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

7. Abbreviations used are defined in 6.1.

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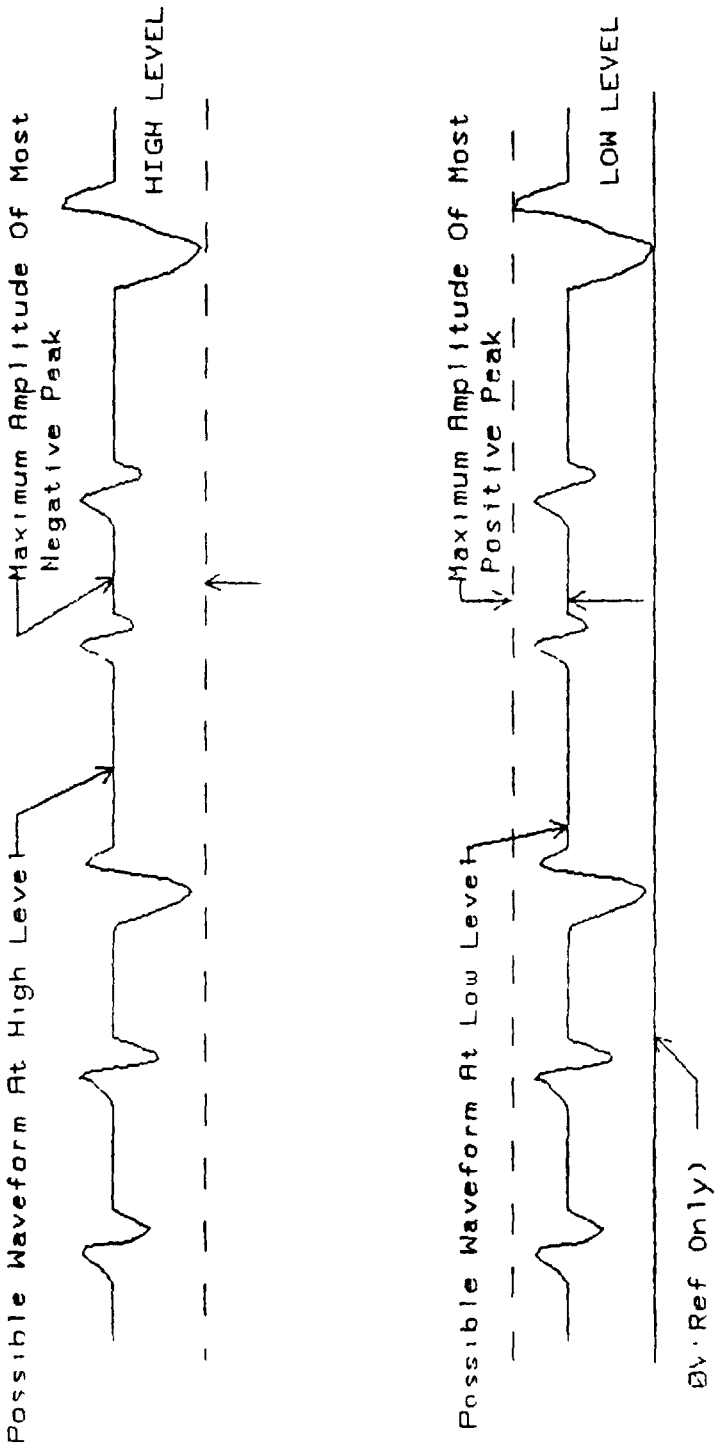


FIGURE 12. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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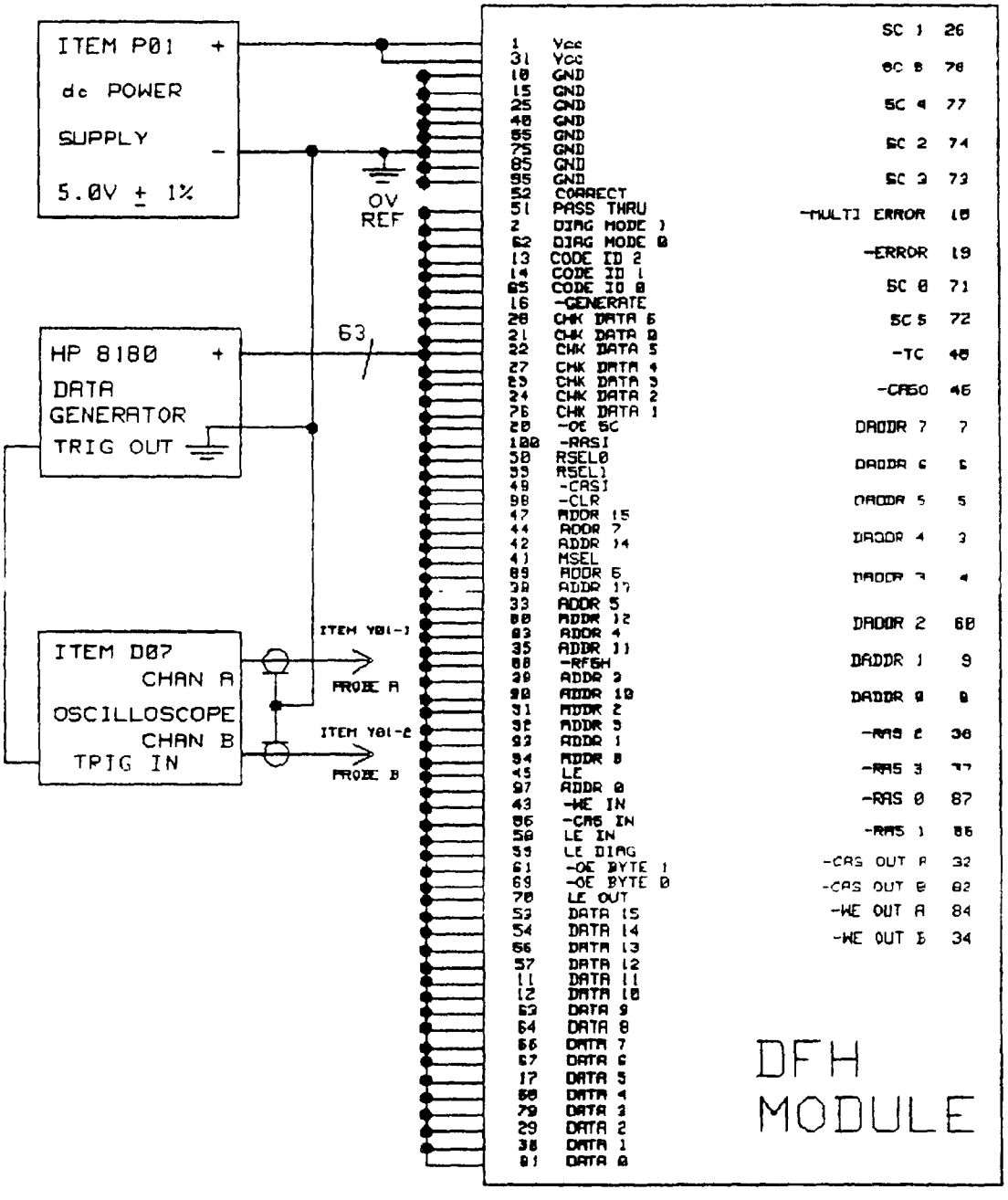


FIGURE 13. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
U1	SC 0	71	0	650	mV	1-7	12,13
U2	SC 1	26					
U3	SC 2	74					
U4	SC 3	73					
U5	SC 4	77					
U6	SC 5	72					
U7	SC 6	78	0	650	mV	1-7	12,13
U8	-TC	48	0	500	mV	1-7	12,13
U9	-CASO	46					
U10	DADDR 0	8					
U11	DADDR 1	9					
U12	DADDR 2	60					
U13	DADDR 3	4					
U14	DADDR 4	3					
U15	DADDR 5	5					
U16	DADDR 6	6					
U17	DADDR 7	7					
U18	-RAS 0	87					
U19	-RAS 1	86					
U20	-RAS 2	36	0	500	mV	1-7	12,13

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs.
U21	-RAS 3	37	0	500	mV	1-7	12,13
U22	-CAS OUT A	32					
U23	-CAS OUT B	82					
U24	-WE OUT A	84					
U25	-WE OUT B	34					
U26	DATA 0	81					
U27	DATA 1	30					
U28	DATA 2	29					
U29	DATA 3	79					
U30	DATA 4	68					
U31	DATA 5	17					
U32	DATA 6	67					
U33	DATA 7	66					
U34	DATA 8	64					
U35	DATA 9	63					
U36	DATA 10	12					
U37	DATA 11	11					
U38	DATA 12	57					
U39	DATA 13	56					
U40	DATA 14	54	0	500	mV	1-7	12,13

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs.
U41	DATA 15	53	0	500	mV	1-7	12,13
U42	-ERROR	19	0	650	mV	1-7	12,13
U43	-MULTI ERROR	18	0	650	mV	1-7	12,13

NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U43
Power supply current limit	1.24 A	NA	U1-U43
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U43 U1-U43
Output crosstalk measurement	 NA NA NA	 ± 32.5 mV ± 25.0 mV ± 32.5 mV	 U1-U7 U8-U41 U42,U43

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 12.

(e) The maximum probe capacitance is 10 pF.

3. The test circuit shall be as indicated on figure 13.

4 All outputs shall be loaded as specified on figure 7.

5. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

6. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

7. Abbreviations used are defined in 6.1.

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TABLE XXII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
V1	DATA 11	11	2	35	pF	1-3
V2	DATA 10	12				
V3	DATA 5	17				
V4	DATA 2	29				
V5	DATA 1	30				
V6	MSEL	41				
V7	DATA 15	53				
V8	DATA 14	54				
V9	DATA 13	56				
V10	DATA 12	57				
V11	DATA 9	63				
V12	DATA 8	64				
V13	DATA 7	66				
V14	DATA 6	67				
V15	DATA 4	68				
V16	DATA 3	79				
V17	DATA 0	81				
V18	-RASI	100				
V19	CODE ID 0	65				
V20	CODE ID 1	14	2	35	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V21	CODE ID 2	13	2	35	pF	1-3
V22	CHK DATA 0	21	2	40	pF	1-3
V23	CHK DATA 5	22				
V24	CHK DATA 3	23				
V25	CHK DATA 2	24				
V26	CHK DATA 4	27				
V27	CHK DATA 6	28	2	40	pF	1-3
V28	-WE IN	43	4	40	pF	1-3
V29	CHK DATA 1	76	2	40	pF	1-3
V30	-CAS IN	96	4	40	pF	1-3
V31	DIAG MODE 1	2	2	35	pF	1-3
V32	-GENERATE	16				
V33	-OE SC	20				
V34	LE	45				
V35	-CASI	49				
V36	RSEL0	50				
V37	PASS THRU	51				
V38	CORRECT	52				
V39	LE IN	58				
V40	LE DIAG	59	2	35	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
V41	-OE BYTE 1	61	2	35	pF	1-3
V42	DIAG MODE 0	62				
V43	-OE BYTE 0	69				
V44	LE OUT	70				
V45	-RFSH	88				
V46	-CLR	98				
V47	RSEL1	99	2	35	pF	1-3
V48	ADDR 5	33	2	40	pF	1-3
V49	ADDR 11	35				
V50	ADDR 13	38				
V51	ADDR 3	39				
V52	ADDR 14	42				
V53	ADDR 7	44				
V54	ADDR 15	47				
V55	ADDR 12	80				
V56	ADDR 4	83				
V57	ADDR 6	89				
V58	ADDR 10	90				
V59	ADDR 2	91				
V60	ADDR 9	92	2	40	pF	1-3

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TABLE XXII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
V61	ADDR 1	93	2	40	pF	1-3
V62	ADDR 8	94	|	|	|	|
V63	ADDR 0	97	2	40	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	V1-V63
Capacitance meter voltage	15 mV	± 1.5 mV	V1-V63
Capacitance measurement	NA	± 1.65 pF	V1-V21
	NA	± 1.9 pF	V22-V27
	NA	± 1.8 pF	V28
	NA	± 1.9 pF	V29
	NA	± 1.8 pF	V30
	NA	± 1.65 pF	V31-V47
	NA	± 1.9 pF	V48-V63

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIII. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W1	SC 1	26	2	40	pF	1-3
W2	SC 0	71				
W3	SC 5	72				
W4	SC 3	73				
W5	SC 2	74				
W6	SC 4	77				
W7	SC 6	78	2	40	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W7
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W7
Capacitance measurement	NA	± 1.9 pF	W1-W7

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Resistance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X1	DADDR 0	8	22	32	Ohm	1-3
X2	DADDR 1	9				
X3	DADDR 2	60				
X4	DADDR 3	4				
X5	DADDR 4	3				
X6	DADDR 5	5				
X7	DADDR 6	6				
X8	DADDR 7	7				
X9	-RAS 0	87				
X10	-RAS 1	86				
X11	-RAS 2	36				
X12	-RAS 3	37				
X13	-CAS OUT A	32				
X14	-CAS OUT B	82				
X15	-WE OUT A	84				
X16	-WE OUT B	34	22	32	Ohm	1-3

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NOTES:

1. Resistance test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply	5.0 V	± 0.25 V	X1-X16
Power supply current limit	1.24 A	NA	X1-X16
Forcing current	5.0 mA 10.0 mA	± 50 uA ± 100 uA	X1-X16 X1-X16
Output voltage measurement	NA	± 16 mV	X1-X16
Pattern voltage low, (VL)	0.0 V	± 100 mV	X1-X16

2. The following procedure is required to perform this test.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot shall not exceed the value specified in the power supply stress test tolerance table.

(b) No load is required for this test.

(c) Measure V and I at 5 mA and 10 mA.

(d) Calculate the output resistance by using the following formula:

$$R_{OUT} = \frac{\Delta V}{\Delta I}$$

3. Abbreviations used are defined in 6.1.

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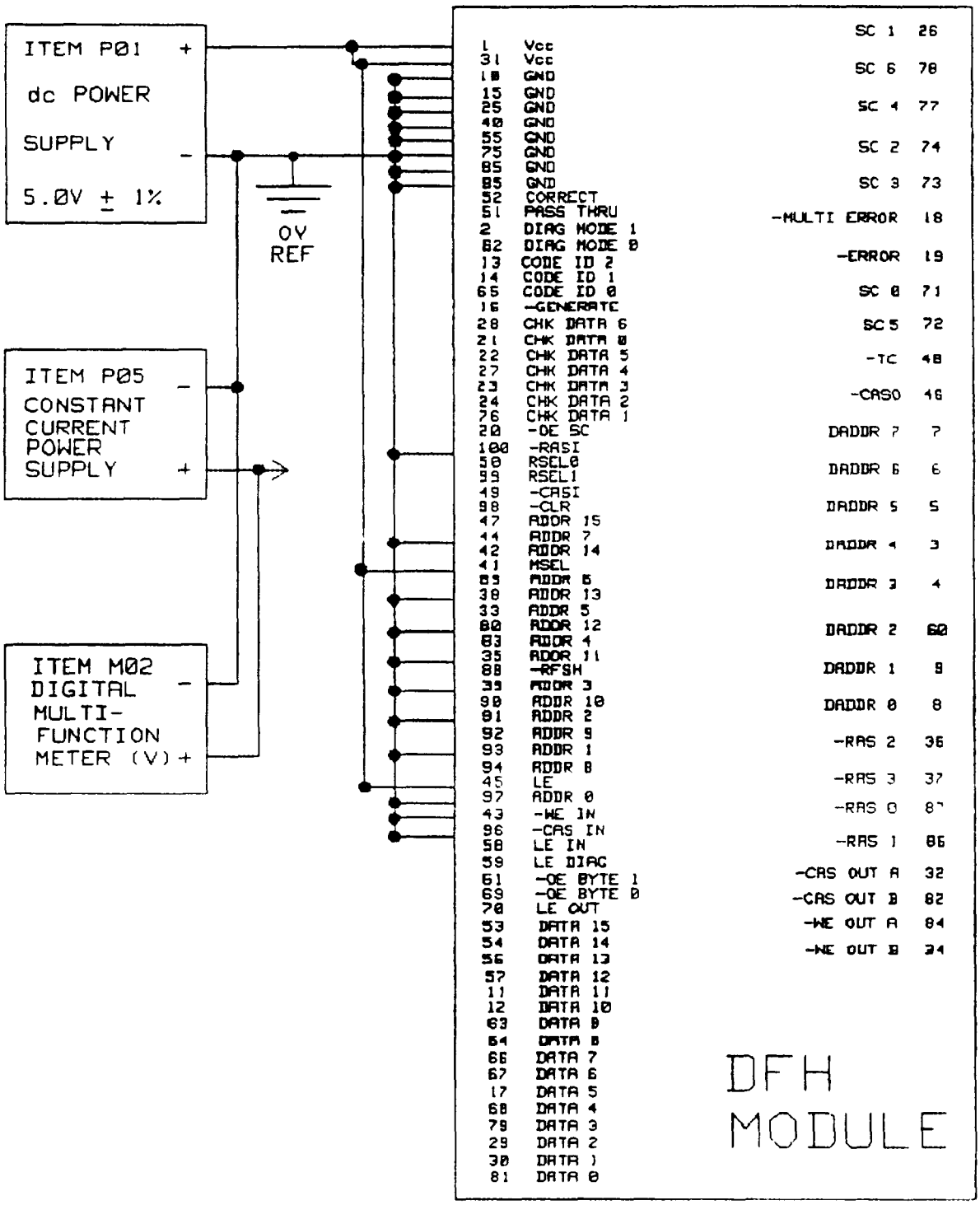


FIGURE 14. Resistance test circuit for table XXIV.

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TABLE XXV. Threshold test.

Test cond	Test description	Notes	Figure
Y	Threshold test	1-4	8

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Y
Power supply current limit	1.24 A	NA	Y
Pattern voltage high (VH) low (VL)	 2.0 V 0.8 V	 ± 20 mV ± 8 mV	 Y Y
Output voltage measurement high (Voh) low (Vol)	 NA NA	 ± 105 mV ± 25 mV	 Y Y

2. The following procedure is required to perform this test.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0 V) or low (0.0 V) during testing.

4. Abbreviations used are defined in 6.1.

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TABLE XXVI. Disable time test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Figs
Low to high impedance (tPLZ)						
Z1	-OE SC to SC 0	71	5	46	nS	15,16
Z2	-OE SC to SC 1	26				
Z3	-OE SC to SC 2	74				
Z4	-OE SC to SC 3	73				
Z5	-OE SC to SC 4	77				
Z6	-OE SC to SC 5	72				
Z7	-OE SC to SC 6	78	5	46	nS	15,16
High to high impedance (tPHZ)						
Z8	-OE SC to SC 0	71	5	90	nS	15,16
Z9	-OE SC to SC 1	26				
Z10	-OE SC to SC 2	74				
Z11	-OE SC to SC 3	73				
Z12	-OE SC to SC 4	77				
Z13	-OE SC to SC 5	72				
Z14	-OE SC to SC 6	78	5	90	nS	15,16
Low to high impedance (tPLZ)						
Z15	-OE BYTE 0 to DATA 0	81	5	46	nS	15,16
Z16	-OE BYTE 0 to DATA 1	30	5	46	nS	15,16

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TABLE XXVI. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z17	-OE BYTE 0 to DATA 2	29	5	46	nS	15,16
Z18	-OE BYTE 0 to DATA 3	79				
Z19	-OE BYTE 0 to DATA 4	68				
Z20	-OE BYTE 0 to DATA 5	17				
Z21	-OE BYTE 0 to DATA 6	67				
Z22	-OE BYTE 0 to DATA 7	66				
Z23	-OE BYTE 1 to DATA 8	64				
Z24	-OE BYTE 1 to DATA 9	63				
Z25	-OE BYTE 1 to DATA 10	12				
Z26	-OE BYTE 1 to DATA 11	11				
Z27	-OE BYTE 1 to DATA 12	57				
Z28	-OE BYTE 1 to DATA 13	56				
Z29	-OE BYTE 1 to DATA 14	54				
Z30	-OE BYTE 1 to DATA 15	53	5	46	nS	15,16

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TABLE XXVI. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
High to high impedance (tPHZ)						
Z31	-OE BYTE 0 to DATA 0	81	5	90	nS	15,16
Z32	-OE BYTE 0 to DATA 1	30				
Z33	-OE BYTE 0 to DATA 2	29				
Z34	-OE BYTE 0 to DATA 3	79				
Z35	-OE BYTE 0 to DATA 4	68				
Z36	-OE BYTE 0 to DATA 5	17				
Z37	-OE BYTE 0 to DATA 6	67				
Z38	-OE BYTE 0 to DATA 7	66				
Z39	-OE BYTE 1 to DATA 8	64				
Z40	-OE BYTE 1 to DATA 9	63				
Z41	-OE BYTE 1 to DATA 10	12				
Z42	-OE BYTE 1 to DATA 11	11				
Z43	-OE BYTE 1 to DATA 12	57	5	90	nS	15,16

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TABLE XXVI. Disable time test - Continued.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z44	-OE BYTE 1 to DATA 13	56	5	90	nS	15,16
Z45	-OE BYTE 1 to DATA 14	54				
Z46	-OE BYTE 1 to DATA 15	53	5	90	nS	15,16

NOTES:

1. Disable time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply	5.0 V	± 0.25 V	Z1-Z46
Power supply current limit	1.24 A	NA	Z1-Z46
Pattern voltage high (VH)	3.0 V	± 100 mV	Z1-Z46
low (VL)	0.0 V	± 100 mV	Z1-Z46
Disable time measurement	NA	± 2.05 nS	Z1-Z7
	NA	± 4.25 nS	Z8-Z14
	NA	± 2.05 nS	Z15-Z30
	NA	± 4.25 nS	Z31-Z46

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NOTES: - Continued.

2. The following procedure is required to perform this test.

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

(d) For each output under test in table XXVI, connect probe 'B' and visually inspect the output for the disable time shown on figure 15.

(e) The maximum probe capacitance is 10pF.

3. The test circuit shall be as indicated on figure 16.

4. All outputs shall be loaded as indicated on figure 7.

5. Floating inputs may cause damage to the module. All inputs shall be tied high (3.0V) or low (0.0V) during testing.

6. Abbreviations used are defined in 6.1.

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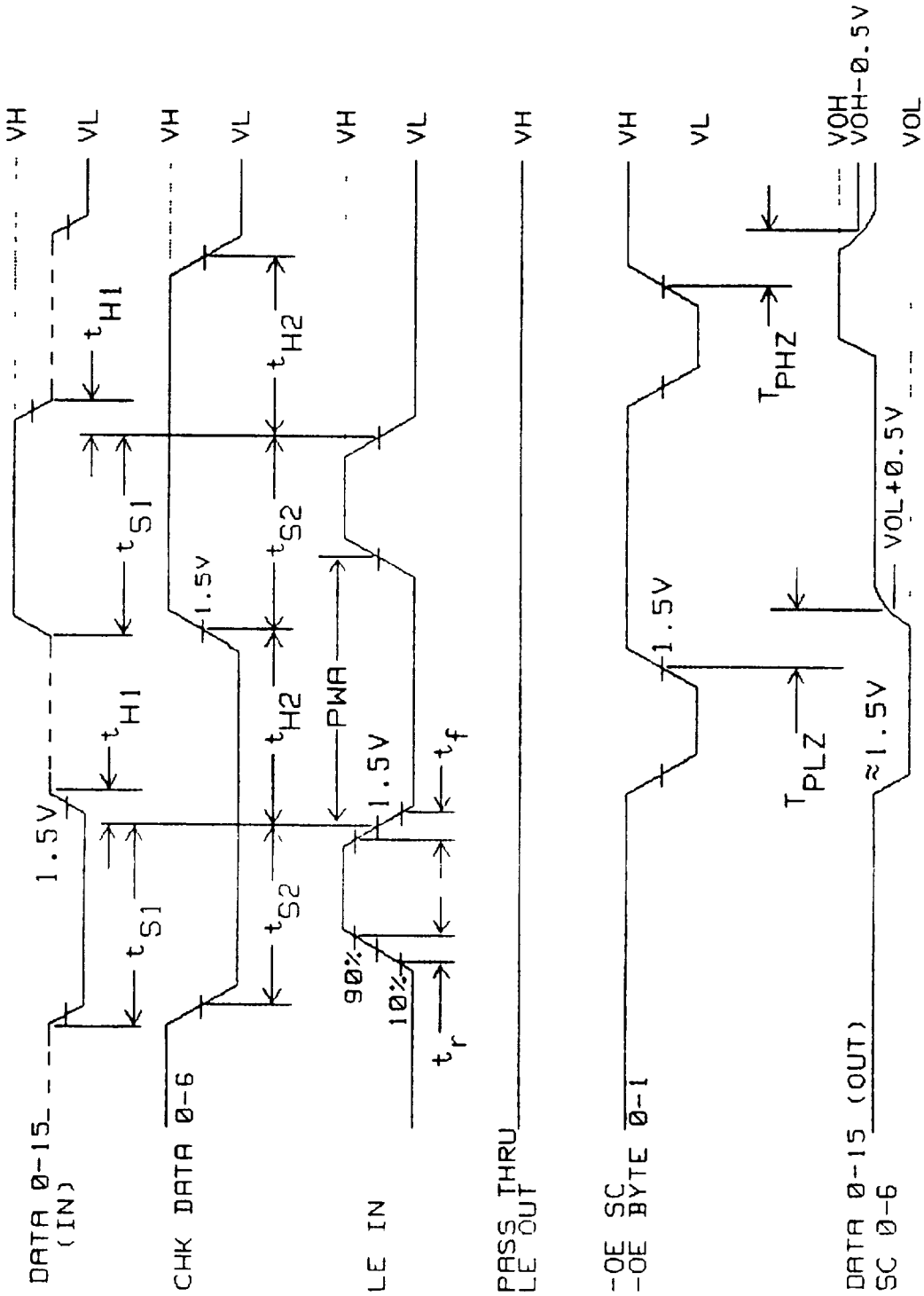


FIGURE 15. Disable time test waveforms for table XXVI.

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NOTES:

1. The input parameters for figure 15 are as follows:

tS1 (DATA0-15) = 7 nS minimum.
tS2 (CHK DATA0-6) = 5 nS minimum.
tH1 (DATA0-15) = 7 nS minimum.
tH2 (CHK DATA0-6) = 7 nS minimum.
PWA (LE IN) = 15 nS minimum.
tr = 15 nS maximum.
tf = 6 nS maximum.

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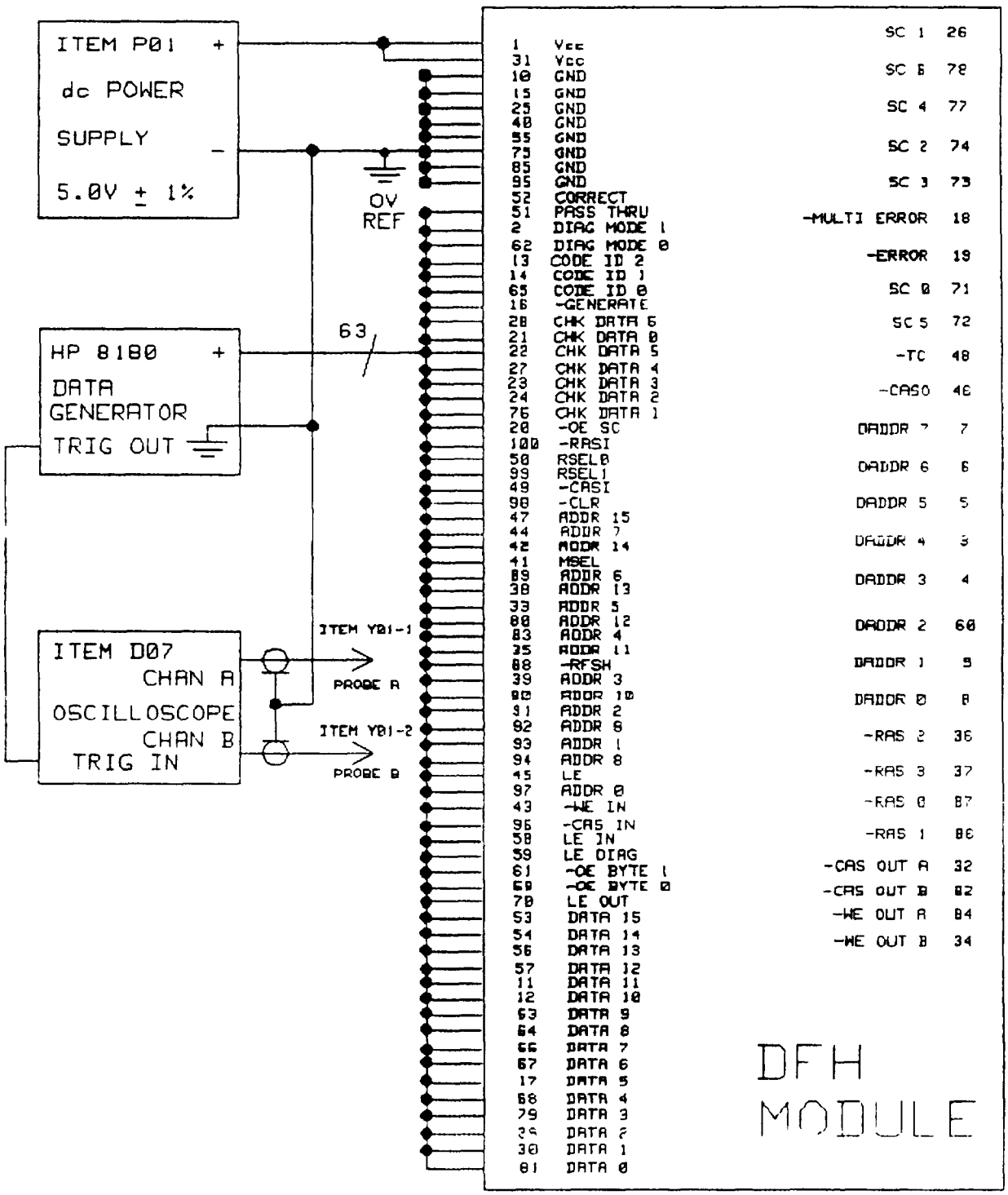


FIGURE 16. Disable time test circuit for table XXVI.

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APPENDIX

MODULE FUNCTION

10. SCOPE

10.1 Scope. This appendix contains information regarding the module function. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

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APPENDIX

30. Module function. The DFH Module decodes memory (DRAM) address, selects the appropriate memory bank and multiplexes the address bits. The module also performs error detection and correction for single bit data errors and detection of double bit errors. The DFH Module provides memory subsystem support in the form of a memory controller (ADC) and error detection and correction (EDC) logic for single bit errors and detection of double bit errors. Figure 1 shows the functional block diagram for the ADC and EDC. The ADC portion of the module provides address decoding to select up to four banks of 64 k words of memory. The ADC also contains memory drivers for driving high capacitance loads and contains series resistors at its outputs used to dampen the undershoot associated with driving DRAM high capacitance inputs. The EDC portion of the module contains built in diagnostic modes which allow the processor to exercise the EDC under software control. The EDC portion of the module controls a standard 16 bit memory word requiring an additional 6 bits for correction memory. The EDC circuitry requires an initialization sequence to bring all memory to a known state. During the initialization sequence, memory is zero filled and the appropriate check bits are placed in corresponding memory locations. There are certain problems to be dealt with when using the EDC in memory sizes other than 16 bit words. Byte write operations require the following steps:

(a) Read the complete data word from memory.
(b) Correct the complete data word if necessary.
(c) Insert the byte to be written into the data word.
(d) Generate new check bits for the entire data word.
(e) Store the data word back into the memory.

Above sequences hold true for any data being written into memory that is not 16 bits wide.

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APPENDIX

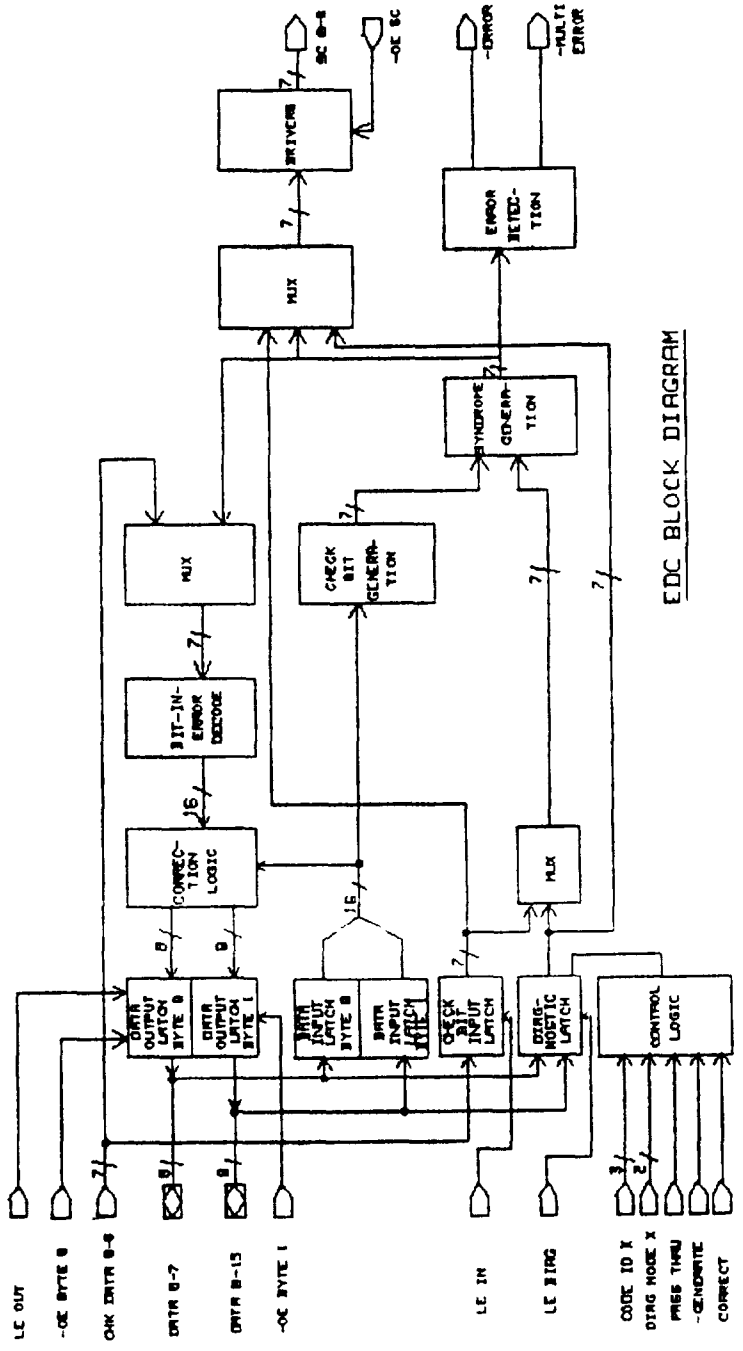


FIGURE 1. DFH module component block diagram.

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APPENDIX

ADC BLOCK DIAGRAM

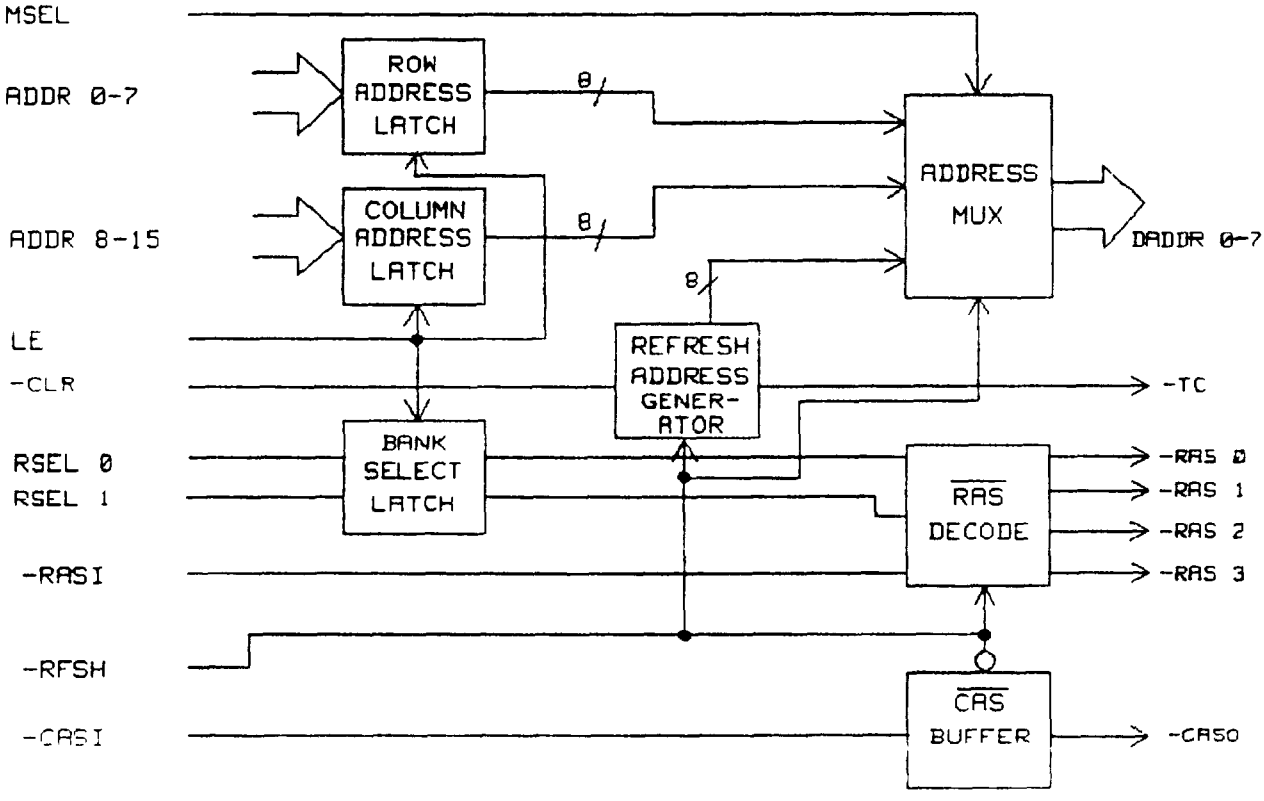


FIGURE 1. DFH module component block diagram - Continued.

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APPENDIX

30.1 Modes of operation.

30.1.1 Error detection/correction function. The EDC portion of the module operates in several modes providing check bit generation detection of errors, correction of errors, pass through mode and diagnostic mode. Additionally, the EDC portion of the module is cascadable such that memory sizes up to 64 bits (72 with correction bits) can be accommodated. It is necessary to indicate to the EDC what size data word is involved and which bytes of the data word it is processing. This is necessary since the EDC operates differently in each configuration. The word size and slice identification is specified through the CODE ID bits.

CODE ID2	CODE ID1	CODE ID0	Hamming Code and Slice Selected
0	0	0	Code 16/22
0	0	1	Internal Control Mode
0	1	0	Code 32/39, Bytes 0&1
0	1	1	Code 32/39, Bytes 2&3
1	0	0	Code 64/72, Bytes 0&1
1	0	1	Code 64/72, Bytes 2&3
1	1	0	Code 64/72, Bytes 4&5
1	1	1	Code 64/72, Bytes 6&7

30.1.2 Control mode selection. The module control lines for the EDC portion of the module are -GENERATE, CORRECT, PASS THRU, DIAG MODE 0-1 and CODE ID 0-2.

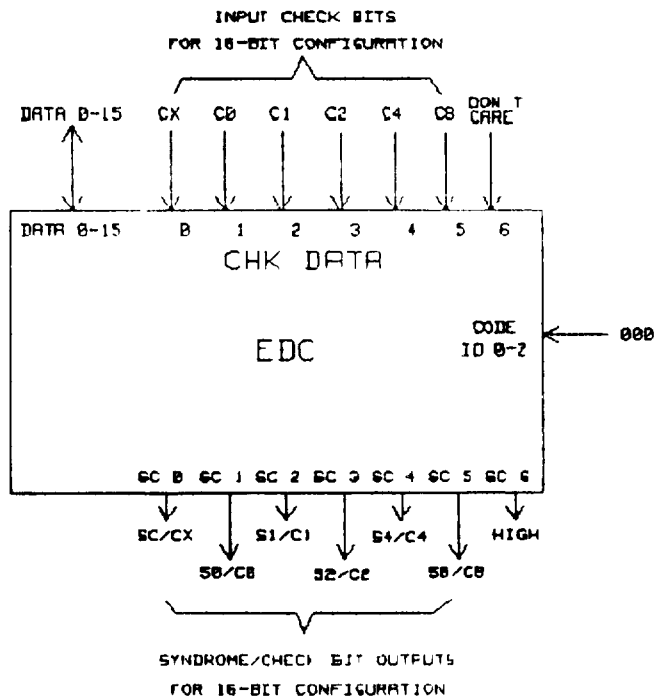
Operating mode	Diagnostic mode**		-Generate	
	DM1	DM0	0	1
Normal	0	0	Generate	Correct*
Diagnostic Generate	0	1	Diagnostic Generate	Correct*
Diagnostic Detect/Correct	1	0	Generate	Diagnostic Correct
Initialize	1	1	Initialize	Initialize
Pass Thru	When PASS THRU is asserted the Operating Mode is defaulted to the Pass Thru Mode.			

*Correct if the CORRECT input is high, detect if the CORRECT input is low.
**In CODE ID 0-2,001 (ID2,ID1,ID0) DM1 and DM0 are taken from the diagnostic latch.

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30.1.3 Generate mode. In this mode the check bits are generated for the corresponding inputs from the memory. The check bits generated are placed on the output lines SC 0 - SC 5 (SC 6 is a logic high). Check bits are generated according to a modified Hamming Code.

30.1.4 Detect mode. In this mode the EDC circuitry examines the data input from memory along with the corresponding check bits and will detect all single bit errors, all double bit errors, and some triple bit errors. If one or more errors are detected, the -ERROR signal goes low. If two or more errors are detected, the -MULTI ERROR signal also goes low. Both of the error indicators remain high when no errors are detected. Also available on the module outputs are the syndrome bits (SC 0-5) generated by the error detection circuitry. The syndrome bits may be decoded to determine if a bit error was detected and, for single bit errors, which of the data bits is in error. If no errors are detected, the syndrome bits remain low. The following figures show the syndrome bit decode to bit error function for 16 bit and 8 bit data word configurations.



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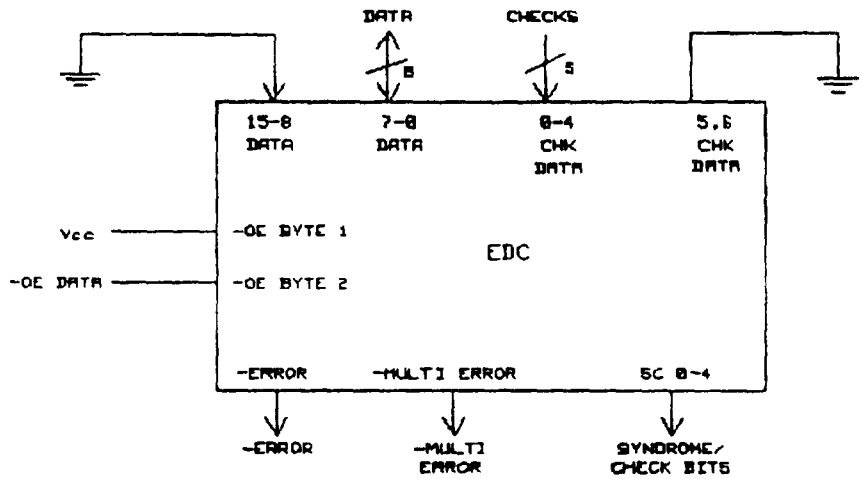
APPENDIX

SYNDROME DECODE TO BIT-IN-ERROR 16-BIT MODE

Syndrome Bits			SC8	0	1	0	1	0	1	0	1
			SC4	0	0	1	1	0	0	1	1
			SC2	0	0	0	0	1	1	1	1
SCX	SC0	SC1									
0	0	0		*	C8	C4	T	C2	T	T	M
0	0	1		C1	T	T	15	T	13	7	T
0	1	0		C0	T	T	M	T	12	6	T
0	1	1		T	10	4	T	0	T	T	M
1	0	0		CX	T	T	14	T	11	5	T
1	0	1		T	9	3	T	M	T	T	M
1	1	0		T	8	2	T	1	T	T	M
1	1	1		M	T	T	M	T	M	M	T

* - no errors detected
T - two errors detected
M - three or more errors detected

Number - the location of the single bit-in-error



SYNDROME DECODE TO BIT-IN-ERROR 8-BIT MODE

SYNDROME BITS		SC4	0	1	0	1
		SC2	0	0	1	1
SC0	SC1					
0	0		*	C4	C2	5
0	1		C1	3	TM	7
1	0		C0	2	1	6
1	1		TM	4	0	TM

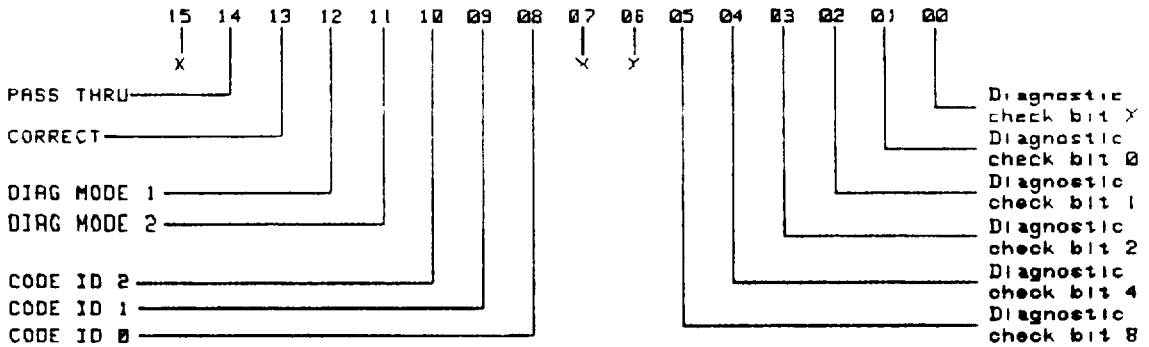
* - no errors detected
TM - two or more errors

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30.1.5 Correct mode. In this mode the EDC circuitry functions the same as in the detect mode, except that the correction circuitry now corrects (complements) any single bit error before outputting the data word. The input data word (taken from the Data Input Latch) is corrected if a single bit error exists, and is placed on the inputs of the Data Output Latch. If multiple errors are detected, the output of the correction network is unspecified. If the single bit error is a check bit, there is no automatic correction. If check bit correction is desired, this can be done by placing the device in the GENERATE mode to produce a correct check bit sequence for the data in the Data Input Latch.

30.1.6 Pass thru mode. In this mode, the unmodified contents of the Data Input Latch are placed on the input of the Data Output Latch, and the contents of the Check Bit Input Latch are placed on the outputs SC 0-5. -ERROR and -MULTI ERROR are forced HIGH in this mode.

30.1.7 Diagnostic mode. The diagnostic latch serves both for diagnostics and internal control use. It is loaded from the DATA lines under control of the LE DIAG signal. Figure 2 shows the loading definitions for the Data lines.



X = Don't care

FIGURE 2. Diagnostic latch load.

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30.1.8 Internal control mode. This mode is selected by inputs at CODE ID 0-2 of 001 respectively. When in Internal Control Mode, the EDC takes the CODE ID 0-1, CORRECT, and PASS THRU control signals from the internal diagnostic latch rather than from the external input lines.

30.2 Dynamic ram address controller function. The Dynamic Address Controller portion of the module provides address multiplexing, refresh address generation, and RAS & CAS control for MOS dynamic RAM memories. Figure 7 contains a block diagram of the AM2964 Dynamic Memory COntroller. The eight bit address path is designed for 64K DRAMs and can be used by 16K DRAMS by pin programming. Sixteen bit address input latches and two RAS select latches (for higher order addresses) allow the controller to control up to 256K words of memory (with 64K DRAMs) by using the internal RAS decoder to select from one-of-four banks of DRAMs.

30.2.1 Input latches. The two eight bit input latches are transparent when LE is HIGH and latch input data meets setup and hold time requirements when LE goes LOW. In systems with separate address and data busses, LE may be permanently enabled HIGH.

30.2.2 Refresh counter. The 8-bit refresh counter provides both 128 and 256 lines refresh capability. Refresh control is external to allow maximum user flexibility. Transparent burst, synchronous or asynchronous refresh modes are all possible. The refresh counter is incremental with the LOW-to-HIGH transition of -RFSH (or -RASI). This assures a stable counter output for the next refresh cycle. The counter will continue to cycle through 256 addresses unless reset to zero by -CLR. This actually causes all outputs to go HIGH since the output MUX is inverting. (Address inputs to outputs are noninverting since both the input latches and output MUX are inverting.)

30.2.3 Terminal count. The refresh counter also provides a terminal count output (-TC) for burst mode refresh applications. -TC normally occurs at count 255 (DADDR 0 to DADDR 7 all LOW when -RFSH is LOW).

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30.2.4 Address multiplexer. The address MUX is 8-bits wide (for 64K DRAMs) and has three data sources, the lower address input latch (ADDR 0 to ADDR 7), the upper address input latch (ADDR 8 to ADDR 15), and the internal refresh counter. The lower address latch is selected when MSEL is HIGH. This is normally the row address. The upper address latch is selected when MSEL is LOW. This is normally the column address. The third source, the refresh counter, is selected when -RFSH is LOW and overrides MSEL. When -RFSH goes LOW, the MUX selects the refresh counter address and -CAS0 is inhibited. Also, the RAS decoder function is changed from one-of-four to four-of-four so all RAS outputs (-RAS 0 to -RAS 3) go LOW to refresh all banks of memory when -RASI goes LOW. WHen -RFSH is HIGH, only one RAS output goes low, determined by the RAS select inputs, RSEL0 and RSEL1. In either case, the RAS decoder output timing is controlled by -RASI to make sure the refresh count appears at DADDR 0 to DADDR 7 before -RAS 0 to -RAS 3 goes low. This assures meeting row address setup time requirements for the DRAM.

30.2.5 -RAS output function.

-RFSH	-RASI	RSEL 1	RSEL 0	-RAS 0	-RAS 1	-RAS 2	-RAS 3
L	H	X	X	H	H	H	H
L	L	X	X	L	L	L	L
H	H	X	X	H	H	H	H
H	L	L	L	L	H	H	H
H	L	L	H	H	L	H	H
H	L	H	L	H	H	L	H
H	L	H	H	H	H	H	L

30.2.6 -CAS0 output function.

-RFSH	-CASI	-CAS0
H	L	L
H	H	H
L	X	H

30.2.7 Address output function.

MSEL	-RFSH	DADDR 0 - DADDR 7
H	H	ADDR 0 - ADDR 7
L	H	ADDR 8 - ADDR 18
X	L	Refresh Address

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30.2.8 Refresh address counter function.

-CLR	-RFSH	-RASI	-TC	REFRESH COUNT	FUNCTION
L	X	X	X	FF H	Clear Counter
H	*F	X	X	NC	Output Refresh Address No Counter Change
H	*R	L	X	Count - 1	Return to Memory Cycle Mode and Decrement Counter
H	L	*F	X	NC	Output all -RAS to RAM No Counter Change
H	L	*R	X	Count - 1	Return -RAS to HIGH and Decrement Counter
H	X	X	L	00 H	Terminal Count for 256 Line Refresh

*R = Rising edge of waveform
*F = Falling edge of waveform
NC = No Count
X = Don't Care

30.2.9 Functional performance of interface circuitry. The interface circuitry for the ADC portion of the module contains drivers which are designed to drive the capacitance input characteristics of the address and control lines of MOS dynamic RAMs. Additionally, this output circuit contains series resistors (6.8 Ohm) used to minimize the undershoot effects associated with driving the address and control lines of MOS dynamic RAMs.

30.2.10 Restrictions. In order to minimize noise pickup from "floating" input pins, all unused inputs should be properly terminated (pulled up to Vcc through pullup resistors or tied to ground).

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